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Masterarbeit

Bayesian Additive Regression Trees for Causal Inference

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Abstract

Bayesian Additive Regression Trees (BART) sind ein verbreitetes Verfahren zur Schätzung durchschnittlicher Kausaleffekte (Average Treatment Effects, ATE). In der bisherigen Forschung wurden verschiedene BART-Meta-Learner allerdings selten systematisch miteinander verglichen und Hyperparameter meist nicht optimiert. Diese Arbeit setzt an diesen beiden Forschungslücken an: Vier BART-Meta-Learner, ein S-Learner, ein T-Learner sowie zwei S-Learner mit geschätzten Propensity Scores als zusätzlicher Kovariate (einmal geschätzt mittels logistischer Regression, einmal mittels BART), wurden mit den beiden Benchmark-Methoden Augmented Inverse Probability Weighting und Causal Forest verglichen. Die Evaluation der Methoden erfolgte anhand von 16 niedrigdimensionalen datengenerierenden Prozessen mit kontinuierlichem Outcome, die von der ACIC 2019 Data Challenge stammen. Die Methoden wurden hinsichtlich der Schätzung des ATE anhand von RMSE, Coverage und Intervalllänge verglichen. Für vier der datengenerierenden Prozesse wurde zusätzlich eine Hyperparameter-Optimierung durchgeführt. Die BART-Modelle schnitten insgesamt besser als die beiden Benchmark-Methoden ab. Der S-Learner erzielte im Durchschnitt den niedrigsten RMSE, während der T-Learner höhere Schätzfehler aufwies, insbesondere bei unbalancierten Daten. Propensity Scores verbesserten die Ergebnisse nur begrenzt, außer wenn der Treatment-Zuweisungsmechanismus komplex war. Geringer Overlap in den Kovariaten verschlechterte die Coverage aller BART-Varianten. Die Hyperparameter-Optimierung wirkte sich nur begrenzt auf die Performance aus und änderte selten, welche Methode am besten abschnitt. Dies spricht dafür, dass die Standardwerte der Hyperparameter bereits nahezu optimal gewählt sind und die Wahl der Methode insgesamt einen größeren Einfluss hat als die Wahl der Hyperparameter. Insgesamt zeigte sich BART, insbesondere der S-Learner, als verlässliche Wahl zur Schätzung durchschnittlicher Kausaleffekte.

Abstract

Bayesian Additive Regression Trees (BART) have become a popular tool for causal effect estimation, but existing studies rarely compare different BART meta-learners systematically, and hyperparameter tuning is mostly neglected. This thesis addresses these gaps by comparing four BART-based meta-learners, an S-learner, a T-learner, and two S-learners incorporating estimated propensity scores (via logistic regression and via BART), against two benchmark methods, augmented inverse probability weighting and a causal forest. All six methods were applied to 16 low-dimensional data generating processes with continuous outcomes from the ACIC 2019 Data Challenge. The models were evaluated on their estimation of the average treatment effect (ATE), using RMSE, coverage, and interval length. Additionally, hyperparameter tuning was performed for a subset of four data generating processes. The BART models generally outperformed the benchmark methods, with the S-learner achieving the highest overall reliability. The T-learner performed worse, particularly under imbalanced data. Including propensity scores helped mainly where treatment assignment appeared complex. Poor covariate overlap reduced coverage across all BART variants. Hyperparameter tuning had only a limited effect on performance and rarely changed which method performed best, suggesting that default hyperparameters are already close to optimal and that the choice of the estimation method matters more than the choice of hyperparameters. Overall, BART, and the S-learner in particular, emerged as a strong and reliable choice for average treatment effect estimation.

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1 Introduction

A central goal across many research fields is to understand the effect that an intervention or treatment has on an outcome. Examples include medicine (e.g. whether a medication leads to an improvement in patient health), policy making (e.g. whether stricter traffic rules lead to fewer accidents) or social sciences (e.g. whether tutoring leads to better test scores). In such settings, the objective is not merely to identify an association between the two variables treatment and outcome. Instead, the goal is to establish and quantify a causal relationship, where the treatment has an effect on the outcome. To do so, researchers typically compare a treatment group, which receives the intervention, with a control group, which does not. As estimating causal effects differs from identifying associations, predictive models alone are generally not sufficient in this context (Holland, 1986). Instead, specific methods which take the distinction between correlation and causation into account are needed.

Over the past decades, a wide range of methods for causal effect estimation have been developed. Recently, machine learning methods have become popular in causal inference as they can model complex relationships between covariates, treatment assignment and outcome and do not require explicit model specification (Feuerriegel et al., 2024). One of those methods is Bayesian Additive Regression Trees (BART), a flexible sum-of-trees model which was originally developed as a predictive model by Chipman et al. (2010) and has been adapted for the purpose of causal inference by Hill (2011). In a variety of studies, causal BART has been shown to yield reliable treatment effect estimates and often performs favorably compared to other methods (Caron et al., 2022; Hahn et al., 2020; Miratrix et al., 2025).

Despite its growing popularity, the literature on causal BART has not yet sufficiently studied two important aspects. First, causal BART can be implemented using different meta-learners. Two common approaches are S-learners and T-learners (Hill, 2011; Künzel et al., 2019). The S-learner estimates treatment effects with a single model for both treated and untreated units. The T-learner uses two separate models for treated and untreated units. Furthermore, propensity scores can be included in the S-learner as an additional covariate (Hahn et al., 2020). In existing studies, often only one model is evaluated while different meta-learners are rarely compared systematically. Second, BART can be finetuned through several hyperparameters which control model complexity and regularization. Existing default values have been established by Chipman et al. (2010). Researchers often adopt those defaults instead of examining whether they are appropriate in different settings. Notably, the original paper which introduced causal BART (Hill, 2011) did not assess whether the established hyperparameters are also optimal for causal inference. Hyperparameter tuning can often substantially affect the performance of causal models (Machlanski et al., 2023). Therefore, neglecting hyperparameter tuning when comparing causal effect estimation models may result in misleading conclusions.

To address these gaps, this thesis investigates the following research questions:

1. How does the performance of different causal BART learners compare?
2. How do BART-based approaches compare to other causal effect estimation methods?
3. To what extent can hyperparameter tuning improve model performance, and can tuning alter which method performs best?

To answer these questions, six causal effect estimation methods are evaluated. Four of these are BART models: an S-learner, a T-learner and two S-learners that include propensity scores. One of them estimates the propensity scores using a logistic regression, the other using a BART model. These models are compared with two other methods: Augmented inverse probability weighting (AIPW) and causal forests. AIPW is a simple parametric yet doubly robust approach, which combines propensity score weighting with an outcome regression model, while causal forests are a tree-based method like BART and specifically designed to detect heterogeneous treatment effects.

The methods are evaluated on 16 synthetic and semi-synthetic data generating processes from the Atlantic Causal Inference Conference (ACIC) Data Challenge of 2019. The analysis focuses on binary treatment assignment, continuous outcomes and average treatment effect estimation. The performance of the models is assessed using the root mean squared error (RMSE), coverage and the length of the confidence or credible intervals, both overall and across different dataset characteristics. Additionally, hyperparameter tuning is performed on four selected DGPs to evaluate the extent to which tuning can significantly improve estimation performance and influence how models perform relative to each other.

This thesis is structured as follows: Chapter 2 introduces BART as a predictive machine learning model. Chapter 3 presents the foundations of causal inference, including the potential outcomes framework and key challenges in estimating causal effects. Chapter 4 reviews general concepts of causal machine learning methods, discusses how BART can be adapted for treatment effect estimation, and introduces the causal forest and AIPW model. Chapter 5 describes the research design, data, and methodology used in the empirical analysis. Chapter 6 presents the empirical results, which are discussed in Chapter 7. Finally, Chapter 8 concludes the thesis.

2 Bayesian Additive Regression Trees

2.1 Classification and Regression Trees

The methodological foundation of Bayesian Additive Regression Trees (BART) are Classification and Regression Trees (CART), introduced by Breiman et al. (1984), which have become a widely used approach for predictive machine learning tasks.

The objective of CART models is to make predictions on an outcome variable Y based on a set of covariates $X = (X_1, X_2, \ldots, X_p)$. X denotes the vector of covariates, while $x = (x_1, x_2, \ldots, x_p)$ denotes a specific realization of X, i.e. the observed covariate values of a single unit. When the outcome variable is categorical, the model is referred to as a classification tree, when it is continuous, it is referred to as a regression tree. In this thesis, BART is used for regression tasks, therefore the following discussion focuses primarily on this setting. The description of the algorithm in this section is based on Hastie et al. (2009, Chapter 4).

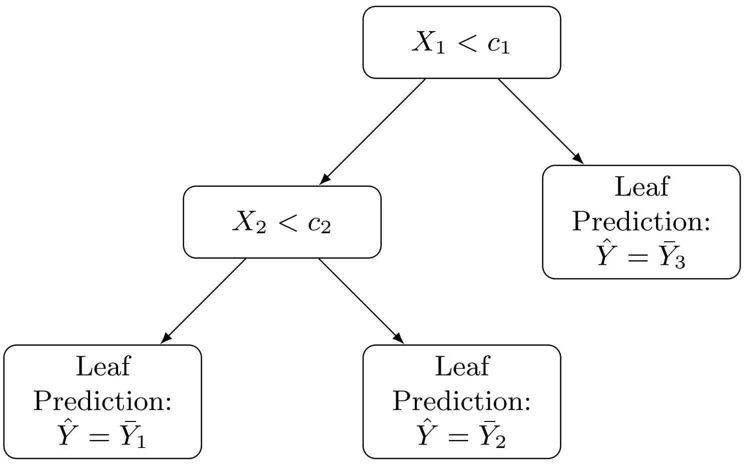


Figure 2.1: Regression tree with two splitting rules and three terminal nodes

A regression tree consists of nodes that are connected by branches, resulting in a hierarchical tree-like structure, displayed in Figure 2.1. The core idea is to recursively partition the covariate space into disjoint regions such that observations within each region are as similar as possible with respect to the outcome variable. The top node is referred to as the root node, while nodes without further splits are called terminal nodes or leaves. The splits are done at the internal nodes based on a splitting rule.

A splitting rule is defined by one of the covariates together with a split value. For numerical covariates, a threshold is chosen. Observations with values smaller than or equal to the threshold are assigned to one branch, while observations with values greater than the threshold are assigned to the other branch. For categorical covariates, categories are partitioned into either of the two branches. To choose a splitting rule, the algorithm compares all different covariates and split values and selects the rule that optimizes a predefined criterion. In regression trees, a common criterion is the residual sum of squares.

Tree construction begins at the root node, using the full training data set. After the first split is selected, the observations are partitioned into two subsets according to the splitting rule. The process is repeated at each of the resulting nodes, considering only the subset of data that was sorted into the node. In principle, this could continue until each observation occupies its own terminal node, resulting in a perfect fit to the training data. Such deep trees typically do not generalize well to new data because they fit the random variation in the training data too closely, which results in low bias but high variance.

To prevent overfitting, it is generally advised to restrict tree growth to a certain depth. Common strategies include defining a minimum number of observations in each node or pruning. In pruning, initially a deep tree is grown and then branches are removed, if they do not improve the predictive performance of the tree sufficiently.

When the tree structure has been finalized, prediction values are assigned to the terminal nodes. Typically, the mean outcome value of the observations in each node is calculated. To make predictions for a new observation, it is passed through the tree according to the splitting rules until it reaches a leaf node. The prediction assigned to that leaf then serves as the predicted outcome.

Regression trees are able to capture nonlinear relationships and complex interactions between covariates without requiring explicit model specification. However, single trees are often unstable, as small changes in the training data can lead to substantially different tree structures. Therefore, ensembles of multiple trees are often used, such as BART.

2.2 Bayesian Additive Regression Trees

Bayesian Additive Regression Trees, first introduced by (Chipman et al., 2010), build upon the concept of CART. The goal remains the same: predicting an outcome Y from a set of covariates X. Unlike standard regression trees, BART is a sum-of-trees model, meaning not one but multiple trees are grown and the final prediction is obtained by summing the predictions of the individual trees. The following description in this section, as well as in Sections 2.3 and 2.4, is based on Chipman et al. (2010).

The model is defined as

$$Y = \sum_{j=1}^{m} g(x; T_j, M_j) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2) \tag{2.1}$$

where m is the number of trees and each function $g(x; T_j, M_j)$ represents a single regression tree. Each tree is defined by its structure T_j, which encapsulates the set of internal nodes with their splitting rules, and a set of terminal node parameters $M_j = \{\mu_{j1}, \mu_{j2}, \ldots, \mu_{jk}\}$. Each μ_{jk} is the prediction assigned to the respective terminal node.

For a given observation with covariates x, the output of each individual tree is the parameter associated with the terminal node reached by x based on the splitting rules. The conditional expectation $E[Y \mid X = x]$ is the sum of the contributions of all trees. The model assumes Gaussian residual variation represented by the error term ε. Thus, the full model can be characterized by the parameters $(T_1, M_1), ..., (T_m, M_m), \sigma$.

BART employs a Bayesian framework by placing priors on the tree structures, the terminal node parameters and the error variance, as well as by using a Gibbs sampler for posterior inference. Together with the data likelihood, this results in a posterior distribution over the model parameters, which in addition to providing point estimates, allows for uncertainty quantification.

Due to its structure, BART is a flexible and robust model. It can naturally handle continuous and categorical covariates. As a nonparametric model, BART does not require assumptions about the functional form of the relationship between the covariates and the outcome. The ensemble of trees is able to capture not only linear relationships, but also interactions and more complex, nonlinear patterns. Regularization is induced by choosing priors, which constrain each individual tree to only explain a small portion of the variation in the outcome. This helps prevent overfitting and leads to more stable predictions. Compared to single tree models, the variance of BART models is typically lower and they are better able to approximate smooth functions. BART has been shown to scale well to high-dimensional settings and is relatively robust to irrelevant covariates, which tend to be selected less frequently during tree construction.

2.3 Prior Specification

Employing a Bayesian framework, BART requires the specification of prior distributions over all model parameters. These parameters include the tree structure T_j, the terminal node parameters M_j and the error variance σ^2. The priors are chosen in such a way that shallow trees are favored, with the goal of regularizing the influence of individual trees. Various prior specifications have been proposed in the literature, and through empirical evaluation default priors have been established. The joint prior distribution over the model parameters is given by

$$p\Big((T_1, M_1), \ldots, (T_m, M_m), \sigma^2\Big) \tag{2.2}$$

Specifying a prior on this joint distribution directly would be impractical. However, the prior can be decomposed into its components. Assuming independence between the trees and independence of σ^2, the joint prior can be written as

$$p\Big((T_1, M_1), \ldots, (T_m, M_m), \sigma^2\Big) = \left[\prod_{j=1}^{m} p(T_j, M_j)\right] p(\sigma^2) \tag{2.3}$$

Applying the product rule for joint distributions yields

$$\left[\prod_{j=1}^{m} p(T_j, M_j)\right] p(\sigma^2) = \left[\prod_{j=1}^{m} p(M_j \mid T_j)\, p(T_j)\right] p(\sigma^2) \tag{2.4}$$

Furthermore, conditional independence between the terminal node parameters within each tree is assumed, resulting in

$$p(M_j \mid T_j) = \prod_{k=1}^{b_j} p(\mu_{jk} \mid T_j) \tag{2.5}$$

where b_j denotes the number of terminal nodes in tree j. Now, instead of placing a prior on the full joint distribution, priors can be placed on $p(T_j)$, $p(\mu_{jk} \mid T_j)$, and $p(\sigma^2)$ separately.

2.3.1 Tree structures

The prior on the tree structure consists of three components. The first component specifies the probability that a node splits at depth d:

$$\Pr(\text{node splits at depth } d) = \alpha(1+d)^{-\beta}, \quad \alpha \in (0,1),\ \beta \in [0, \infty) \tag{2.6}$$

where α is the base split probability and β is the rate at which the splitting probability decreases with depth. A higher α encourages more splits, meaning the tree will grow larger. Through a higher β, this probability decreases more quickly with increasing depth. Chipman et al. (2010) recommend $\alpha = 0.95$ and $\beta = 2$. These values encourage splits near the root and cause the splitting probability to decrease rapidly, which favors shallow trees overall.

The second component is the probability that a covariate X is selected for a splitting rule at a node. The default prior is a discrete uniform distribution over all covariates. The third component specifies the prior distribution over the splitting values of each covariate X. This prior is usually a uniform distribution over the observed available split points of X.

2.3.2 Terminal node parameters

Typically, Y is rescaled to approximately lie within the interval $[-0.5, 0.5]$. This is not mathematically necessary, but facilitates prior specification and interpretation. The prior distribution for each terminal node parameter is then given by

$$\mu_{jk} \sim \mathcal{N}(0, \sigma_\mu^2) \tag{2.7}$$

with

$$\sigma_\mu = \frac{0.5}{k\sqrt{m}} \tag{2.8}$$

where m is the number of trees and k is a shrinkage parameter. The constant 0.5 encourages the sum of the tree contributions to remain within a range compatible with the rescaled outcome. The prior aims to shrink the terminal node parameters toward zero, which leads each tree to contribute only a small portion of the variation in the outcome. Both a higher number of trees m and higher values of the shrinkage parameter k result in stronger shrinkage toward zero. The recommended default parameter by Chipman et al. (2010) for k is 2. m is usually set to at least 50, while 100 or 200 are also common choices (Tan & Roy, 2019).

2.3.3 Error variance

Another prior is placed on the error variance σ^2. In the original BART specification, σ^2 follows an inverse-chi-square distribution:

$$\sigma^2 \sim \frac{\nu\lambda}{\chi^2_\nu} \tag{2.9}$$

where ν denotes the degrees of freedom and λ is a scale parameter. ν controls the shape of the distribution and the heaviness of its tails. Smaller values of ν result in heavier tails, allowing for more flexible variance estimates. λ determines the scale of the prior distribution and is not fixed directly. It is chosen in a data-driven way based on the empirical variance of the outcome variable, $\hat{\sigma}_y^2$, such that

$$P(\sigma < \hat{\sigma}_y) = q, \tag{2.10}$$

where q is the fraction of prior mass placed below the empirical standard deviation of the outcome $\hat{\sigma}_y$. ν and q are often specified as a pair. Typical values recommended by Chipman et al. (2010) are the conservative specification $(10, 0.75)$, which places more prior mass on smaller variances, the more aggressive specification $(3, 0.99)$, which allows for much larger variances, and $(3, 0.90)$, which avoids extremes and is often used by default.

2.4 Markov Chain Monte Carlo Algorithm

The posterior over the BART parameters (the tree structures T_j, the terminal node parameters M_j, and the error variance σ^2) is estimated with a Markov Chain Monte Carlo (MCMC) algorithm. Sampling from the joint posterior of $p\Big((T_1, M_1), \ldots, (T_m, M_m), \sigma^2 \mid Y, X\Big)$ directly is intractable. Therefore, a Gibbs sampler is implemented which iteratively samples from the full conditional distributions. Since it is not possible to sample directly from the conditional distribution of the tree structures, a Metropolis-Hastings step is embedded within the Gibbs sampler, resulting in a Metropolis-within-Gibbs algorithm. The terminal node parameters and the error variance can be drawn in closed form from their full conditional distributions.

The algorithm follows a Bayesian backfitting scheme, where each tree is updated conditional on all the other trees. Instead of fitting each tree to the outcome Y, each tree is fitted to the partial residuals, which represent the variation in the outcome not explained by the other trees.

One iteration of the Gibbs sampler proceeds as follows:

First, the partial residuals are calculated as

$$R_j = Y - \sum_{l \neq j} g(X; T_l, M_l) \tag{2.11}$$

Conditional on R_j, a new tree structure T_j^* is proposed by making a small random change to the current tree structure T_j. The options for modifying the tree typically are:

- grow: split one of the current terminal nodes,
- prune: remove an existing split,
- change: modify the splitting rule of an internal node,
- swap: exchange the splitting rules of two internal nodes.

The type of modification is randomly chosen. A proposal distribution assigns a probability to each of the modifications. Common implementations place higher probability on grow and prune moves than on change and swap moves. In case of a grow move, the covariate and the splitting value are selected according to the priors defined in Section 2.3.1.

To decide whether to accept the new tree T_j^* or keep the current tree T_j, the Metropolis-Hastings acceptance ratio is calculated.

$$\alpha = \min\left(1, \frac{p(R_j \mid T_j^*)\, p(T_j^*)\, q(T_j \mid T_j^*)}{p(R_j \mid T_j)\, p(T_j)\, q(T_j^* \mid T_j)}\right) \tag{2.12}$$

This ratio compares whether the proposed tree structure fits the data better than the old tree structure. It consists of three components.

The first component is the likelihood ratio

$$\frac{p(R_j \mid T_j^*)}{p(R_j \mid T_j)} \tag{2.13}$$

which measures whether the proposed tree improves the fit to the partial residuals compared to the current tree. The likelihood

$$p(R_j \mid T_j) = \int p(R_j \mid T_j, M_j)\, p(M_j \mid T_j)\, dM_j \tag{2.14}$$

is the marginal likelihood of the tree. The terminal node parameters M_j are integrated out analytically, which is possible because conjugate priors are used for the terminal node parameters. Using the marginal likelihood instead of conditioning on M_j improves the efficiency of the MCMC algorithm.

The second component

$$\frac{p(T_j^*)}{p(T_j)} \tag{2.15}$$

is the ratio of the priors of the two tree structures. It acts as a penalty for complexity, favoring shallow trees. Larger trees should only be accepted if they improve the fit enough to justify additional splits.

The third component is the proposal ratio

$$\frac{q(T_j \mid T_j^*)}{q(T_j^* \mid T_j)} \tag{2.16}$$

which adjusts for the fact that some modifications are more likely than others (e.g. a large tree offers more opportunities for pruning than a small one). This correction ensures that the Markov chain has the posterior as its invariant distribution.

If the proposed tree is accepted, T_j is replaced by T_j^* . Otherwise, T_j is kept.

Conditional on the updated tree structure, the terminal node parameters are sampled from their full conditional distributions

$$p(\mu_{jk} \mid T_j, R_j) \tag{2.17}$$

which are normally distributed. After all m trees have been updated, the error variance σ^2 is sampled from its full conditional distribution.

The above procedure describes one draw from the posterior distribution. At the beginning, the trees are initialized consisting only of a root node. The sampling procedure is then repeated many times, disregarding a specified number of burn-in samples to reduce the influence of the starting values. After a sufficient number of iterations, the generated draws approximate samples from the posterior distribution. These posterior draws can then be used to generate predictions for new data and to construct Bayesian credible intervals.

3 Causality

3.1 Potential Outcomes Framework

To formally define causal effects throughout this thesis, the potential outcomes framework (POF) will be used. As it was introduced by Neyman (1990) (originally published 1923) and extended by Rubin (1974) to observational studies, it is also known as the Neyman-Rubin causal model (NRCM).

A population or sample consists of n units indexed by $i = 1, \ldots, n$. For each unit, X_i denotes the vector of observed covariates, Y_i denotes the outcome variable, and Z_i denotes the treatment assignment, indicating whether the unit receives treatment or not. In the case of binary treatment assignment, $Z_i = 1$ indicates that the unit receives treatment, $Z_i = 0$ indicates that the unit does not. For each unit, there are two potential outcomes: $Y_i(1)$ is the outcome that would be observed if the unit received treatment, $Y_i(0)$ is the outcome that would be observed if the unit did not receive treatment (Rubin, 1974).

To illustrate the concepts, the effect of tutoring on students' test scores will be used as an example throughout this thesis. Here, the units are the students, Z_i indicates whether a student attends tutoring classes, and Y_i is the student's test score. $Y_i(1)$ would be the test score of student i if they attended tutoring classes. $Y_i(0)$ would be the test score if they did not.

Based on the potential outcomes, several estimands can be defined. The individual treatment effect (ITE) is the causal effect that the treatment has on an individual unit and is defined as the difference between the two potential outcomes of that unit:

$$\tau_i = Y_i(1) - Y_i(0) \tag{3.1}$$

Since the two potential outcomes refer to the same unit, they differ only in treatment status. Therefore, any difference observed between the potential outcomes can be attributed to the treatment.

This calculation is, however, purely conceptual, because in reality a unit either receives treatment or it does not. Therefore, only one of the outcomes can be observed, while the other remains unobserved and is referred to as the counterfactual outcome. This is known as the fundamental problem of causal inference. Since both potential outcomes cannot be observed simultaneously for the same unit, the individual treatment effect cannot be directly identified. Because of that, causal inference typically focuses on aggregate measures of causal effects.

The average treatment effect (ATE) is the average causal effect of the treatment across the entire population:

$$ATE = \tau = E[Y(1) - Y(0)] \tag{3.2}$$

The sample average treatment effect (SATE) can be written as the average of the individual treatment effects:

$$SATE = \tau_S = \frac{1}{n}\sum_{i=1}^{n}\Big(Y_i(1) - Y_i(0)\Big) \tag{3.3}$$

Treatment effects can differ across subpopulations. In the tutoring example, tutoring could work better for students from certain socio-economic backgrounds. When treatment effects are evaluated conditional on covariates, the conditional average treatment effect (CATE) is obtained:

$$CATE = \tau(x) = E[Y(1) - Y(0) \mid X = x] \tag{3.4}$$

If treatment effects vary across units depending on their covariates, the treatment effect is referred to as heterogeneous. If the treatment effect is identical for all units, it is referred to as homogeneous. In this case, the CATE is equal to the ATE for all covariate values.

Two additional important estimands are the average treatment effect on the treated (ATT), which measures the average causal effect among units that actually received treatment:

$$ATT = E[Y(1) - Y(0) \mid Z = 1] \tag{3.5}$$

and the average treatment effect on the controls (ATC):

$$ATC = E[Y(1) - Y(0) \mid Z = 0] \tag{3.6}$$

Similar to the individual treatment effect, average treatment effects cannot be directly observed. However, under certain assumptions, they can be estimated through comparisons between treatment and control group. In observational settings, differences between these groups may not only be caused by the treatment itself, but also by additional variables that influence the outcome and treatment assignment. These variables are referred to as confounders.

3.2 Confounders

A confounder is a variable that has an influence on both the outcome and on treatment assignment (Pearl, 2009). This creates a problem because the observed association between treatment and outcome cannot be attributed solely to the treatment. Instead, some or even all of the observed association may be driven by the confounder, which affects both variables. As a result, estimates of treatment effects may be biased. The relationship between confounders, treatment and outcome is displayed in Figure 3.1.

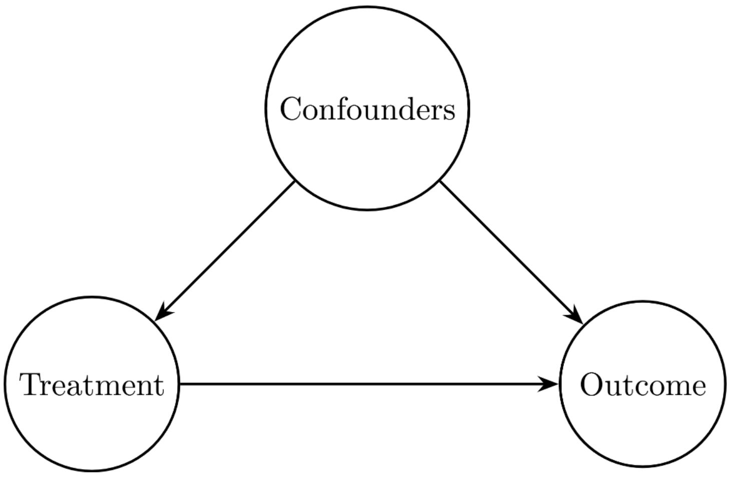


Figure 3.1: Relationship between confounders, treatment assignment and outcome

In the tutoring example, a confounder could be previous academic performance. Students who previously had lower grades are typically more likely to attend tutoring classes. They are also more likely to get a lower test score. Previous grades therefore have a relationship with both the treatment and the outcome. Ignoring this confounder could lead to misleading conclusions about the effectiveness of tutoring. In this case, it could falsely suggest that attending tutoring classes leads to lower test scores, even if the actual causal effect is positive.

To still obtain valid estimates of causal effects, it is necessary to control for confounding variables (Pearl, 2009). The goal is to remove the part of the association between the treatment assignment and the outcome, which the confounders induce. Then, only the effect of the treatment on the outcome remains. To identify relevant confounders, domain knowledge is often required (Chen et al., 2024).

3.3 Randomized controlled trials

The standard approach of estimating causal effects in scientific research is a randomized controlled trial (RCT) (Shadish et al., 2002). Here, participants are divided into two groups. One group, called the treatment or experimental group, receives the treatment, the other, called the control group, does not. An important aspect of RCTs is that the assignment to either treatment or control group happens completely at random. Since there are no variables that influence who is assigned to which group, the groups are on average comparable in terms of observed and unobserved variables. As a result, systematic differences in the outcome variable are assumed to be caused by the treatment, not by other covariates (Stolberg et al., 2004). Apart from controlling for known confounders, randomization also controls for unknown confounders already at the design stage of the study.

While RCTs are considered the gold standard for causal inference, they are not always feasible. Conducting them can them be expensive or impossible (Shadish et al., 2002). Therefore, observational studies are very common in practice.

3.4 Observational studies

Observational studies are studies where the assignment of the treatment is not under the control of the researcher (Imbens & Rubin, 2015). While RCTs are conducted with the explicit goal of investigating a specific research question, observational studies often use data that was collected for other purposes rather than for the study itself. This typically makes observational studies less costly and more flexible in scope (Shadish et al., 2002).

In the tutoring example, the data may not come from a controlled experiment but instead from a school database with student information. The database records test scores and whether a student attended tutoring. The decision to assign a student to tutoring was made by the parents or the teacher rather than through random assignment. Treatment assignment is therefore likely influenced by covariates such as prior academic performance or socio-economic background. This induces systematic differences between the treatment and control group, leading to confounding (Rosenbaum & Rubin, 1983). In contrast to RCTs, which control for confounders through randomization at the design stage, observational studies must address confounding in the analysis stage (Imbens & Rubin, 2015).

3.5 Assumptions for Causal Inference

To estimate causal effects from non-randomized data, several assumptions are required.

3.5.1 Ignorability Assumption

The ignorability assumption, also referred to as unconfoundedness assumption, assumes that there are no unobserved confounders (Rubin, 1974). In order to obtain unbiased estimates of treatment effects, all relevant confounders must be observed and controlled for. The assumption can be formally written as (Rosenbaum & Rubin, 1983):

$$Y(0), Y(1) \perp Z \mid X \tag{3.7}$$

This means that, conditional on the observed covariates X, treatment assignment Z is independent of the potential outcomes. In other words, once controlled for X, there are no other factors that influence treatment assignment and outcome. This assumption cannot be tested directly, since by definition, unobserved confounders cannot be observed.

Previous grades as a possible confounder in the tutoring example were already discussed in Section 3.2. If previous grades are not observed and controlled for, the ignorability assumption is violated and estimates may be biased.

3.5.2 Overlap Assumption

The overlap assumption, also referred to as common support assumption, requires that for every covariate pattern there is a positive probability of both receiving and not receiving treatment (Rosenbaum & Rubin, 1983):

$$0 < P(Z = 1 \mid X = x) < 1 \quad \text{for all } x \tag{3.8}$$

That means that there are no subgroups of individuals who always receive or who never receive treatment. The assumption is important because treatment effects can only be estimated when treated and untreated individuals with comparable covariate values exist.

In the tutoring example, the assumption would be violated if all students from a certain socio-economic background were assigned to tutoring and not a single person from this subgroup would remain in the control group.

3.5.3 Stable Unit Treatment Value Assumption

The stable unit treatment value assumption (SUTVA) assumes that each unit has only one potential outcome under each treatment condition (Rubin, 1980). This assumption is required to ensure the stability and interpretability of causal effects.

Two conditions need to be fulfilled for SUTVA to hold:

1. There is no interference between units. This means that the potential outcomes of one unit are not affected by the treatment status of another unit. In the tutoring example, the assumption would be violated if students attending tutoring shared learning material with students in the control group, influencing the test scores of the control group.

2. There are no different variations of treatment. All treated units must receive the same version or intensity of the treatment. In the tutoring example, this would require all students in the treatment group to attend the same number and type of tutoring sessions.

4 Causal Inference Methods

4.1 From Predictive to Causal Machine Learning

Standard machine learning models aim to predict outcomes accurately, causal inference on the other hand focuses on the estimation of treatment effects. Predictive associations do not necessarily imply causal relationships, therefore predictive machine learning methods cannot automatically be applied to causal tasks (Shmueli, 2010).

A variety of methods have been developed for the estimation of treatment effects (a review of current research trends can be found in Arti et al. (2020)). These methods have different approaches in addressing confounders, either by trying to create a population which mimics an RCT or by modelling the underlying data generating processes. In general, there are two mechanisms that are of interest (Imbens & Rubin, 2015). One is the treatment assignment mechanism, which describes the process determining which individuals receive treatment and which do not, represented by $Z \mid X$. The second is the outcome mechanism, which captures how the covariates and the treatment determine the outcome together, represented by $Y \mid Z, X$. Many causal machine learning methods estimate these mechanisms using so-called nuisance models (Chernozhukov et al., 2018). These are models which are not of primary interest, but are used to control for confounding and improve treatment effect estimation.

In this chapter, a short introduction to the methods used throughout this thesis will be provided.

4.1.1 Implications for Model Evaluation

Due to the differences between predictive and causal models, two things need to be considered when evaluating causal inference methods.

Full-Sample Estimation

Predictive models aim to predict the outcome of new observations. A good model should not only learn the relationship in the data it is trained on, but be able to generalize well to unseen data. Therefore, when evaluating predictive models, data is typically split into training and test data (Stone, 1974). The model is fit to the training data and evaluated on the test data.

In contrast, in causal inference the goal is not primarily outcome prediction, but the estimation of causal parameters, mainly the treatment effect. This task is complicated by the fact that the ground truth is never observed (Athey & Imbens, 2016). Because standard out-of-sample validation cannot be applied in this setting, causal effect estimation is often performed using the full dataset in order to make use of all available information.

Data Sources for Evaluating Causal Effect Estimators

The difference in evaluation strategy extends further: Because standard predictive validation cannot confirm whether a causal estimate is correct, researchers additionally face the challenge of how to evaluate causal estimators at all. The best option seems to be real-world data, as it ensures that the results are relevant for practical applications. However, this leads to the fundamental problem of causal inference: The true treatment effect is not known, since for each unit, only one potential outcome can be observed at a time (Holland, 1986). As a result, model performance cannot be evaluated directly.

For this reason, synthetic data is often used. In this case, the covariates, the treatment assignment mechanism and the outcome mechanism are simulated. The ground truth is known and model estimates can be directly compared to true treatment effects. Additionally, key characteristics of the data, such as overlap or the degree of treatment effect heterogeneity, can be systematically varied.

Synthetic data has some clear limitations: It is often less complex than real-world data, therefore results from simulation studies may not transfer well to practical problems (Curth et al., 2021). Another issue with simulated data is that researchers can choose data generating processes which favor some methods over others, especially if they align with the assumptions of those methods.

To address these issues, semi-synthetic data is often favored over synthetic data (Neal et al., 2020). Here, the covariates and in some cases the treatment assignment come from real-world data. Only the outcome model and sometimes treatment assignment is simulated. This introduces some of the complexity of real data, while still allowing for a known ground truth.

4.2 Propensity Scores

A widely used tool in causal inference is the propensity score. The propensity score is the probability of receiving treatment conditional on the covariates (Rosenbaum & Rubin, 1983):

$$e(X) = P(Z = 1 \mid X) \tag{4.1}$$

It captures the relationship between the covariates and treatment assignment and can therefore be used to control for confounding. An advantage of propensity scores is that they can reduce a high-dimensional set of covariates to a single variable (Olmos & Govindasamy, 2015).

Propensity scores are typically estimated using logistic regression or other binary classification models (Lee et al., 2010; Rosenbaum & Rubin, 1983). While propensity scores are not a standalone method for treatment effect estimation, they are often incorporated into other methods. An example is propensity score matching, where each observation from the treatment group is matched to one or more observations from the control group with similar propensity scores and only those observations are included in further analysis models (Rosenbaum & Rubin, 1983). This way, differences in covariate distributions between treatment and control group, which are introduced by confounders, are reduced.

4.3 Meta-learners

A common approach to estimating heterogeneous treatment effects is to adapt standard predictive machine learning models using so-called meta-learners. A meta-learner is a framework in which one or more predictive models can be embedded as base learners (Künzel et al., 2019). It defines a strategy for how these models can be used to estimate treatment effects.

The S-learner (single learner) uses a single model, in which the treatment indicator is included as an additional covariate (Künzel et al., 2019):

$$\hat{\mu}(x, z) = E[Y \mid X = x, Z = z] \tag{4.2}$$

For estimating the conditional average treatment effect for an observation, predictions are made twice for each observation. While keeping all other covariates fixed, the treatment indicator is set once to $Z = 1$ and once to $Z = 0$. The difference between these predictions is the estimate of the conditional average treatment effect:

$$\hat{\tau}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0) \tag{4.3}$$

The T-learner (two learners) fits two models separately, one using the observations in the treatment group and one using the observations in the control group (Künzel et al., 2019):

$$\begin{aligned} \hat{\mu}_1(x) &= E[Y \mid X = x, Z = 1] \\ \hat{\mu}_0(x) &= E[Y \mid X = x, Z = 0] \end{aligned} \tag{4.4}$$

Each model then generates predictions for all observations, their difference estimates the conditional average treatment effect:

$$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x) \tag{4.5}$$

Both S- and T-learners are fitted using the observed outcomes. For the estimation of treatment effects, they predict both the factual and the counterfactual outcome.

There exist other meta-learners like X-learners or R-learners (Künzel et al., 2019; Nie & Wager, 2021). This thesis focuses only on S- and T-learners.

4.3.1 Strengths and limitations of meta-learners

A limitation of the T-learner is that it can only use a subset of the data for each of its models, which can be especially problematic if the dataset is small or imbalanced (when one group contains substantially more observations than the other) (Künzel et al., 2019). In these settings, the S-learner has an advantage, since it uses information from all observations for a single model. This is particularly beneficial when treatment effects are relatively homogeneous.

The S-learner may, however, struggle when the conditional outcome functions $\mu_1(x)$ and $\mu_0(x)$ differ substantially, i.e. when treatment effects are highly heterogeneous (Zhang et al., 2025). The T-learner can better account for differences between the groups because it fits two separate models.

4.3.2 Effects of Regularization

Two further limitations arise for S-learners when they are used in combination with regularized predictive models. In high-dimensional settings, the inclusion of many covariates can reduce bias but may also increase variance (R. Tibshirani, 1996). Therefore, regularization is used to only include covariates in the model that have strong predictive power, i.e. by penalizing a high number of covariates. Since S-learners treat treatment assignment like any other covariate, regularization may shrink the treatment effect towards zero or exclude treatment assignment from the model (Künzel et al., 2019).

A regularized model may also weight down or exclude confounders that have a weak relationship with the outcome but a strong relationship with treatment assignment. As a result, confounding is not sufficiently controlled for, leading to biased treatment effect estimates. This phenomenon is referred to as regularization-induced confounding (Hahn et al., 2018).

4.4 BART for Causal Inference

As BART showed strong predictive performance due to its flexible non-parametric structure, Hill (2011) proposed its use for causal inference. In this original approach, BART is applied as an S-learner. The sum-of-trees model can then be written as

$$Y = f(X, Z) + \varepsilon \tag{4.6}$$

with

$$f(X, Z) = \sum_{j=1}^{m} g(X, Z; T_j, M_j) \tag{4.7}$$

where m is the number of trees, Z denotes treatment assignment, and each function $g(\cdot)$ represents a single regression tree with tree structure T_j and corresponding terminal node parameters M_j.

Later work by Künzel et al. (2019) implemented BART as a T-learner. Hahn et al. (2020) introduced another variation to address the issue of regularization-induced confounding. In BART, this issue can arise because the priors encourage shallow trees and shrink the terminal node parameters toward zero. To address this, they recommended including an estimate of the propensity score as an additional covariate in the model, resulting in

$$Y = f(X, Z, \hat{e}(X)) + \varepsilon, \tag{4.8}$$

where $\hat{e}(X)$ denotes the estimated propensity score.

This way, information about treatment assignment is explicitly included in the model. The propensity score can be estimated using different methods, such as logistic regression or BART itself.

Since its introduction, the basic causal BART model has been extended to a wide range of settings, among these are continuous or multi-categorical treatments (Souto & Louzada Neto, 2024) or time-to-event outcomes (Kabata et al., 2026). These extensions are beyond the scope of this thesis, as they are not part of the empirical evaluation.

4.5 Benchmark methods

In addition to BART, this thesis considers two established benchmark methods for estimating treatment effects: augmented inverse probability weighting (AIPW) and causal forests. Unlike causal BART, neither method is based on the meta-learner framework. Instead, both target the causal estimand directly.

4.5.1 Augmented Inverse Probability Weighting

AIPW combines a model for treatment assignment with a model for the outcome mechanism to improve robustness against model misspecification, an idea by Robins et al. (1994) originating in semiparametric missing-data theory and later adapted to causal inference by Bang & Robins (2005). Its treatment assignment component is based on inverse probability of treatment weighting (IPTW) by Robins et al. (2000), which aims to mimic random treatment assignment by creating a pseudo-population in which treatment is independent of the observed confounders. This is achieved by estimating propensity scores for each observation and weighting the observations with the inverse probability of the treatment they actually received.

For treated observations, Robins et al. (2000) define the weights as

$$w_i = \frac{1}{e(X_i)} \tag{4.9}$$

and for control observations as

$$w_i = \frac{1}{1 - e(X_i)} \tag{4.10}$$

This way, observations that have a low probability of being in their respective group receive a higher weight and those who have a high probability receive a lower weight. As a result, overrepresented observations are downweighted and underrepresented observations are upweighted, which leads to more balanced covariate distributions between the groups (Chesnaye et al., 2022; Robins et al., 2000).

When propensity scores are close to 0 or 1, the weights become extreme (Cole & Hernán, 2008). This can result in a few observations influencing the result disproportionately and increased variance. To reduce this problem, stabilized weights are often used, where the numerator 1 in Equations 4.9 and 4.10 is replaced with the marginal treatment probability, i.e. the proportion of treated (or untreated) units in the full sample.

Since IPTW uses propensity scores as weights, results can vary depending on how well the propensity score model captures the true treatment assignment model. To increase robustness against model misspecification, augmented inverse probability weighting (AIPW) additionally incorporates a model for the outcome mechanism (Robins et al., 1994). Instead of directly using the outcome model to predict the treatment effect, the weighted outcomes are included in the predictions through an augmentation term. The estimator of the ATE can be written as

$$\hat{\tau}_{AIPW} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{Z_iY_i - (Z_i - \hat{e}(X_i))\hat{\mu}_1(X_i)}{\hat{e}(X_i)} - \frac{(1-Z_i)Y_i + (Z_i - \hat{e}(X_i))\hat{\mu}_0(X_i)}{1-\hat{e}(X_i)}\right] \tag{4.11}$$

where $\hat{\mu}_1(X_i)$ and $\hat{\mu}_0(X_i)$ denote the predicted outcomes under treatment and control from the outcome model. Methods that combine a treatment assignment model and an outcome model this way are referred to as *doubly robust* (Scharfstein et al., 1999). Their strength lies in that even if one of the models is misspecified, they can still produce consistent estimates.

Weighting causes some observations to contribute more strongly to the pseudo-population than others (Robins et al., 2000). Therefore, when calculating standard errors, the weighting procedure must be taken into account. Typically bootstrap methods are used to estimate standard errors.

4.5.2 Causal forest

Causal forests are an ensemble tree-based approach for causal effect estimation. All descriptions in this section are based on the paper from Wager & Athey (2018), who developed the method. As discussed in Section 2.1, in regression trees used for prediction, splitting rules are chosen so that observations within the same terminal node have similar outcomes. In contrast, causal forests aim to detect heterogeneity in treatment effects. Splits are selected to maximize differences in treatment effects between terminal nodes. The goal is to separate observations with different treatment effects and to group observations with similar treatment effects together. Within each terminal node, treatment effects are estimated by comparing the outcomes of treated and control observations.

Randomness is introduced to avoid that all trees in the ensemble have the same structure. Each tree is built on only a random subsample of the data and at each internal node, only a random subset of the covariates is considered for determining the splitting rule. Additionally, causal forests typically estimate nuisance models for the treatment assignment mechanism and the outcome mechanism. These models are used to remove variation explained by the covariates before estimating treatment effect heterogeneity. This process is referred to as local centering.

To reduce overfitting, causal forests commonly employ so-called *honest trees*. The data is split into two parts, one subsample is used for determining the tree structure, the other is used to estimate the treatment effects within the terminal nodes. This separation reduces bias in treatment effect estimation.

Hyperparameters

Wager & Athey (2018) also introduced several hyperparameters which can be tuned. First, there is the number of trees in the forest. A larger number of trees generally reduces the variance of the estimator, however, with more trees computational costs increase.

Another important hyperparameter is the number of covariates that are considered when determining the splitting rule at an internal node. Lower values reduce correlation between trees, but considering too few covariates may result in important covariates being excluded from the splitting process too often.

The minimum node size determines how many observations a terminal node must at least contain, therefore controlling when splitting stops. Lower values allow the causal forest to detect more heterogeneous treatment effects, but very small terminal nodes can increase the risk of overfitting.

Another tunable parameter is the subsample fraction, i.e. the proportion of observations used to build each tree. Larger subsamples may reduce variance but can increase similarity between the trees.

Additional hyperparameters include the proportion of data used for splitting rules versus estimation under honesty, whether empty terminal nodes should be pruned, the maximum allowed imbalance of a split and penalties for imbalanced splits. These parameters mainly affect the bias-variance trade-off and the stability of treatment effect estimation.

4.6 Empirical Performance of BART and Benchmark Methods

In general, causal BART models have shown strong empirical performance across a wide range of studies and datasets. They often outperformed AIPW estimators (Degtiar & Finucane, 2023; Kim et al., 2023; McConnell & Lindner, 2019) as well as causal forests and other tree-based methods in terms of bias and coverage (Caron et al., 2022; Hahn et al., 2020; Lu et al., 2018; Miratrix et al., 2025; Wendling et al., 2018).

When comparing different meta-learners in combination with BART, results are mixed and appear to be dependent on the underlying data generating process. Some studies found that S-learners achieve lower RMSE than T-learners across a wide range of data settings (Hahn et al., 2020; Miratrix et al., 2025). Other studies report advantages for the T-learner in settings with more complex treatment effect heterogeneity. For example, Caron et al. (2022) found that T-learners outperformed S-learners on a dataset, where the outcome functions for treatment and control group differed substantially. In contrast, S-learners performed better in a second simulation setting with less complex treatment effect functions. Similarly, Künzel et al. (2019) found that S-learners performed better than T-learners in small, imbalanced samples, however, they used a random forest as base learner instead of BART. Overall, the literature suggests that no single meta-learner consistently dominates across all settings.

Regarding the inclusion of propensity scores, Hahn et al. (2020) found that a propensity score augmented S-learner achieved lower bias and improved coverage compared to standard BART across several simulated datasets, although the improvements were smaller than the differences between S- and T-learners. McConnell & Lindner (2019) similarly reported lower bias for BART when propensity scores were estimated using logistic regression. In contrast, Wendling et al. (2018) found only modest and inconsistent improvements of propensity score-augmented BART over standard BART in simulated settings for heterogeneous treatment effect estimation.

5 Methodology

5.1 Research Goal

Despite the general strong performance of BART models for causal effect estimation, results are not always clear. This is especially true, when it comes to the comparison of different BART learners to each other. Additionally, hyperparameter tuning is usually neglected completely. Therefore, the goal of this thesis is to provide a systematic comparison of causal BART learners, both with each other and with alternative methods. Three research questions will be evaluated:

1. How does the performance of different causal BART learners compare?

2. How do BART-based approaches compare to other causal effect estimation methods?

3. To what extent can hyperparameter tuning improve model performance, and can tuning alter which method performs best?

5.2 Data Source

As discussed in Section 4.1.1, real-world data does not allow for the evaluation of causal estimators against a known ground truth. In this thesis, the objective is to compare models in terms of how well they estimate treatment effects. This requires data for which the true treatment effect is known. At the same time, simulating data is avoided, as this could impose data generating processes that may favor certain models. On the other hand, it is important to compare model performance across different data settings to be able to assess robustness under varying circumstances. Therefore, (semi-)synthetic data generated by external sources is used, ensuring that the underlying data generating processes are not specified within this thesis. Among available benchmark datasets, the choice fell on data from the Atlantic Causal Inference Conference (ACIC) 2019 Data Challenge (Gruber et al., 2019) as it provides a diverse collection of datasets generated under different data generating processes.

The ACIC Data Challenge is a yearly competition held at the Atlantic Causal Inference Conference since 2016, which focuses on solving challenges in causal inference (Dorie et al., 2019). Datasets containing covariates, treatment assignments and outcomes are provided by the organizers and participants are tasked with estimating different causal parameters. The participants can submit models of their choice to solve the tasks. After the challenge closes, the true treatment effects are disclosed and the winning models are announced.

The datasets have become popular beyond the data challenge and are often used as benchmark datasets in causal inference research (e.g. (Mahajan et al., 2022; Ramadhanti et al., 2025; Wang et al., 2026)). They offer the opportunity to evaluate methods on a range of datasets generated independently of the researchers.

5.3 Overview of the Data Generating Processes

In the 2019 challenge, the task was to provide point estimates of the ATE as well as 95% confidence intervals. 64 different DGPs were provided, with 100 datasets generated for each DGP. All datasets featured binary treatment assignment. There were two different tracks, one for low-dimensional datasets and one for high-dimensional datasets. Within each track, both binary and continuous outcomes were included.

For this thesis, only the low-dimensional DGPs with continuous outcomes were selected, resulting in 16 DGPs or 1600 datasets. Table 5.1 provides an overview of the selected datasets. The original names of the DGPs are retained, therefore DGP identifiers are not consecutive. The number of covariates ranges from 22 to 31, the number of observations from 500 to 5735.

For some DGPs, all covariates were simulated, for others the covariates were taken from real-world datasets. In all cases, the treatment assignment mechanism and outcome mechanism were simulated. The assumption of unconfoundedness was satisfied for all datasets. The organizers stated that the datasets were designed to reflect a variety of challenging data characteristics, but it was not disclosed which specific characteristics were present in each DGP (Gruber et al., 2019). This makes the datasets suitable for evaluating the robustness of causal inference methods across a range of unknown data generating conditions.

Table 5.1: Characteristics of the data generating processes

DGP	Number of Covariates	Number of Observations	True ATE
29	26	5735	4
30	26	5735	10
31	26	5735	2
32	26	5735	2.527
37	22	500	4
38	22	500	2.5
39	22	500	6.27
40	22	500	6.962
57	31	649	0.5
58	31	649	1
59	31	649	-0.8
60	31	649	1.686
61	29	649	2
62	29	649	2
63	29	649	-2
64	29	649	3.625

5.4 Data Characteristic Measures

The two measures calculated to characterize the data were balance and overlap. Both measures were calculated across 5 datasets per DGP and results were averaged.

The balance can be directly calculated as the proportion of observations in the treatment group in relation to the total number of observations.

The overlap is the degree to which treatment and control group share similar covariate distributions (see Section 3.5). Exactly quantifying overlap is not straightforward in high-dimensional settings. Common estimation approaches include comparing the propensity score distributions between treatment and control groups or identifying the proportion of observations with extreme propensity scores (Austin, 2011). These methods rely on correct propensity score estimation and do not account for the fact that not all covariates contribute equally to the outcome mechanism (Hill & Su, 2013). A method based on propensity scores might identify insufficient overlap for a covariate that is irrelevant for the outcome and is therefore not a genuine confounder.

Hill & Su (2013) therefore proposed an alternative approach based on the posterior draws of BART. The underlying idea is that in regions with good overlap, the model has enough information from both the treatment and control group and can make stable estimates of individual treatment effects. In regions with poor overlap, the model must extrapolate to generate counterfactual predictions, which leads to increased uncertainty of those predictions. Therefore, observations that do not have a counterfactual in the same covariate region can be identified by comparing the variance of their predicted counterfactual outcomes to the variance of the factual outcome predictions. If the variance of the counterfactual predictions is substantially higher, this serves as an indicator that there is no adequate counterfactual in that covariate region.

The method was used to estimate the level of overlap in the datasets. Based on a rule defined by Hill & Su (2013), observations were flagged if their counterfactual outcome standard deviation s_i^{1-a} exceeded the maximum factual standard deviation across all observations in the same treatment group, plus one standard deviation of that factual distribution:

$$s_i^{1-a} > m_a + \mathrm{sd}(s_j^a) \tag{5.1}$$

where $m_a = \max_j\{s_j^a\}$ for all j with $Z_j = a$, and $\mathrm{sd}(s_j^a)$ denotes the standard deviation of the factual outcome standard deviations across all units in the same treatment group.

5.5 Experimental Procedure

Four different BART meta-learners were compared to assess whether their methodological differences lead to differences in estimation performance. The four meta-learners were an S-learner, a T-learner and two S-learners which include estimates of the propensity scores. One estimated propensity scores using a logistic regression and is referred to as PS-GLM, the other estimated the propensity scores using a BART model and is referred to as PS-BART.

To place the performance of BART in context, the BART models were also compared with two established causal inference methods. First, AIPW was included as a parametric, doubly robust benchmark. Second, causal forests were included as a widely used non-parametric method which, similar to BART, is based on trees.

The empirical evaluation consisted of two stages. Both stages were repeated across all DGPs, allowing an assessment of how performance varies under different data conditions.

In the first stage, all models were estimated using the default hyperparameters provided by the respective software packages. This allowed a first evaluation of the performance differences. Furthermore, this may likely be a setting which researchers use often.

In the second stage, hyperparameters were tuned on a subset of the DGPs for each model. This way, it could be investigated whether model performance can be improved under optimal hyperparameter settings and whether the ranking of methods can be changed by tuning.

5.6 Evaluation Metrics

To evaluate the performance of the models, several metrics are computed for each data generating process.

The root mean squared error (RMSE) measures the estimation accuracy of the ATE:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{\tau}_i - \tau)^2} \tag{5.2}$$

where i is the index of the dataset within the DGP, $\hat{\tau}_i$ is the estimated ATE from dataset i of the DGP, τ is the DGP's true ATE, and $n = 100$ is the number of datasets per DGP. Lower RMSE values indicate greater estimation accuracy.

To assess uncertainty, at each iteration 95% uncertainty intervals are constructed and it is recorded whether the interval contains the true ATE. The interpretation of these intervals differs between frequentist and Bayesian methods. For the frequentist approaches causal forests and AIPW, confidence intervals are reported. If the ATE estimation were repeated and confidence intervals were constructed many times, 95% of these confidence intervals would be expected to contain the true ATE. In contrast, Bayesian methods, such as BART, construct credible intervals, which allow the statement that the true ATE lies within the credible interval with a posterior probability of 95%, conditional on the observed data. For each model and DGP, the average interval length and the coverage rate, defined as the proportion of intervals containing the true ATE, are calculated. Ideally, the empirical coverage rate of a 95 % confidence interval should be close to 0.95.

Since the DGPs differ in the magnitude of the true ATE, RMSE and confidence interval length are not directly comparable across DGPs. When the true ATE is large, DGPs may exhibit a disproportionally large RMSE or interval lengths. This does not affect comparisons between models within a single DGP but can distort results when averaging across all DGPs. To address this issue, RMSE and average interval length were normalized by the true ATE of the respective DGP. Coverage is not affected by scaling and was therefore not normalized.

Not only the average performance across DGPs is of interest, but also the relative performance of the models. In practical applications, researchers usually select a single method. Therefore, it was evaluated how often each model performed best. The models were ranked within each DGP for each metric and summarized statistics were reported across all DGPs.

5.7 Model Implementation

All analyses were conducted in R (v4.6.0; R Core Team, 2026).

The BART T-learner was implemented using the `dbarts` package (v0.9-33; Dorie et al., 2026). This package uses discrete trees, meaning the covariate values for the splitting rules are selected from a discrete set of covariate values, not a continuous one. The `bart2` function from the package was used because it allows for the specification of hyperparameters. Two separate BART models were fitted, one to the treatment group and one to the control group. Predictions were then made on both models for all observations. ITE estimates were calculated as the difference between the predicted treated and control outcomes.

The BART S-learner, PS-BART and PS-GLM were implemented using the bartc function from the `bartCause` package (v1.0-10; Dorie & Hill, 2020). The package internally uses `dbarts`, so differences in performance cannot be attributed to different implementations. For all three models, a single model was fitted using the covariates and treatment assignment as input. The propensity score variants additionally included estimated propensity score estimates as covariates. Propensity scores were estimated within the bartc function. For PS-GLM, they were estimated using logistic regression, for PS-BART, a separate BART model was fitted with treatment assignment as outcome. ITE estimates were obtained by predicting the potential outcomes under treatment and control and calculating their difference.

For all BART models, the number of chains was set to 10, with 500 burn-in samples and 500 posterior samples per chain. These settings correspond to the default values of the `bartCause` package and are consistent with recommendations by Dorie et al. (2019) and Carnegie (2019).

Causal forests were implemented using the `causal_forest` function from the `grf` package (v2.6.1; J. Tibshirani et al., 2026).

For AIPW, the augmented inverse probability weighting estimator was implemented using the `AIPW` package (v0.6.9.3; Zhong et al., 2021). Both the outcome regression and the propensity score model were estimated via the `SuperLearner` package (v2.0.40; van der Laan et al., 2007) using a logistic model for treatment assignment and a linear regression for the outcome. Propensity scores were truncated at the 0.1 bound to avoid extreme weights.

5.8 Default Hyperparameters

The default hyperparameters of the `dbarts` package were used for all BART models: $k = 2, m = 75,\ \alpha = 0.95,\ \beta = 2,\ (\nu, q) = (3, 0.90)$.

For the causal forest, the default hyperparameters from the `grf` package were *num.trees* = 2000, *min.node.size* = 5, *sample.fraction* = 0.5, *honesty.fraction* = 0.5, $\alpha = 0.05$, *honesty.prune.leaves* = *TRUE* and *imbalance.penalty* = 0. *mtry* (number of variables considered at each split) was set dynamically within the package, as $\sqrt{p} + 20$, where p denotes the number of covariates.

5.9 Hyperparameter Tuning

BART models and causal forests both contain a number of hyperparameters. Due to limited computational resources, not all hyperparameters were tuned. Instead, only the parameters for which literature expects a substantial impact on results, were tuned, while the others were kept at their default values. Hyperparameter tuning was conducted using a grid search.

Tuning of the BART models was based on the tuning in Chipman et al. (2010) and focused on the parameters governing the prior distribution of the terminal node values (k and m) and the error variance (nu and q). For k, the values (1, 2, 3, 5) and for m the values (75, 100, 200) were evaluated. For the variance prior, the combinations (3, 0.90), (3, 0.99), and (10, 0.75) were considered. This resulted in a $4 \times 3 \times 3$ grid with a total of 36 combinations.

For the causal forest, tuning was based on Dandl et al. (2024) and focused on the parameters *mtry* (the number of covariates that are considered for chosing the splitting rule), *min.node.size* (minimum number of observations in a terminal node) and *sample.fraction* (proportion of observations used to build each tree). For *min.node.size*, values were (5, 10, 20, 50) and for sample.fraction (0.3, 0.5). Higher values are not possible as at least half of the sample is needed for calculation of standard errors. *mtry* was set dynamically based on the number of covariates in the dataset: $n_{cov}/2$, $n_{cov}/3$, and $\min(\sqrt{n_{cov}} + 20,\ n_{cov})$, with all values rounded up to the nearest integer. This resulted in a $4 \times 2 \times 3$ grid with a total of 24 combinations.

AIPW does not contain hyperparameters in the classical sense. However, both the propensity score model and the outcome model can be estimated through different methods. A larger number of possible model specifications exists, including interaction terms and higher-order polynomials. Exploring all these possible models was beyond the scope of this thesis. Therefore, an alternative, more flexible method for propensity score estimation was evaluated. BART is not easily implemented using the `AIPW` package. Therefore, the underlying logic of the AIPW package was recreated using the `dbarts` package for the treatment assignment mechanism. The propensity score truncation bound (0.1) and the absence of cross-fitting were kept identical across both implementations to ensure comparability.

For hyperparameter tuning, four DGPs were selected based on the results from the first stage of the experiment. The chosen DGPs had relatively big performance differences between the methods, so that potential improvements due to hyperparameter tuning could have a big impact. Additionally, they covered the full range of dataset sizes present in the study, both regarding the number of observations and the number of covariates. To reduce computational costs, only the first 20 datasets were considered for tuning. For each dataset, all methods were run with all the specified hyperparameter combinations. For each combination, the RMSE was calculated. The hyperparameter combination with the lowest RMSE was chosen for each method. Since RMSE is reported to three decimal places in the tables, some combinations may appear tied even though the underlying selection was based on the unrounded value. Then, the RMSE from the tuned method and the RMSE from the default parameters were compared. For the default parameters, only the 20 first datasets were considered, so that they matched the datasets used for tuning.

6 Results

6.1 Default Hyperparameters

6.1.1 Aggregated Results

The following results provide an aggregated overview of the performance of the methods across all DGPs. Table 6.1 shows the mean normalized RMSE, normalized interval length and the coverage of each method, rounded to the third decimal. Figures .1 to .3 show boxplots of the metrics to provide an additional information on how the metrics are distributed across the DGPs. Note that the horizontal line in each boxplot marks the median, so it may differ from the mean values.

The S-learner had the lowest mean RMSE (0.091), closely followed by PS-GLM (0.092) and PS-BART (0.093). The mean RMSE of the T-learner was somewhat higher (0.099), while the causal forest reached 0.117 and AIPW has the highest RMSE of 0.171.

For coverage, the three S-learner-based BART models performed best again. Note that, in contrast to RMSE, higher coverage values are preferred. PS-BART had the highest coverage (0.939), followed by PS-GLM (0.936) and the S-Learner (0.933). The causal forest had a coverage of 0.913, the T-Learner of 0.893 and AIPW had the lowest coverage of 0.858.

Interval length roughly followed a similar pattern to RMSE. The S-Learner produced the shortest intervals (0.357), followed by PS-GLM (0.363), the T-learner (0.366) and PS-BART (0.367). AIPW (0.384) and particularly, the causal forest (0.415) had the longest intervals.

Standard deviations across the 16 DGPs show that AIPW was the least stable model, with the highest variability in both RMSE (SD = 0.278) and coverage (SD = 0.236). The T-learner had the lowest RMSE variability (SD = 0.043) but the second-highest coverage variability (SD = 0.102). The three BART S-learners and the causal forest showed the most stable coverage (SDs between 0.046 and 0.076), while interval length variability was similar across all models (SDs 0.198–0.210).

Table 6.1: Mean RMSE, coverage and interval length by model

Model	RMSE	Coverage	Interval Length
BART/PS-BART	0.093	0.939	0.367
BART/PS-GLM	0.092	0.936	0.363
BART/S-learner	0.091	0.933	0.357
BART/T-learner	0.099	0.893	0.366
Causal forest	0.117	0.913	0.415
AIPW	0.171	0.858	0.366

6.1.2 Performance Across Data Generating Processes

The metrics vary heavily between models as well as between DGPs. This section systematically examines the performance of each method across the 16 DGPs. Figures 6.1 to 6.3 show the normalized RMSE, coverage and normalized interval length for each method across all 16 DGPs. The exact numbers can be found in Tables .1 to .4. A summary of how models ranked relative to each other across the 16 DGPs can be found in Tables .5 to .7.

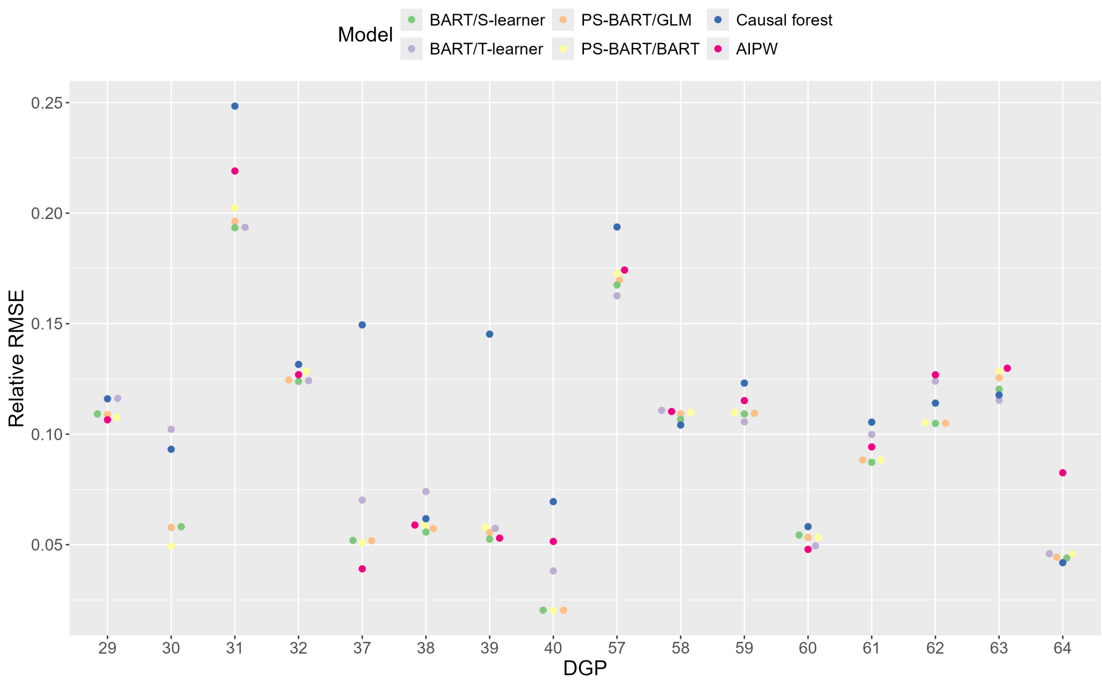


Figure 6.1: Relative RMSE by model and DGP (AIPW's outlier in DGP 30 excluded).

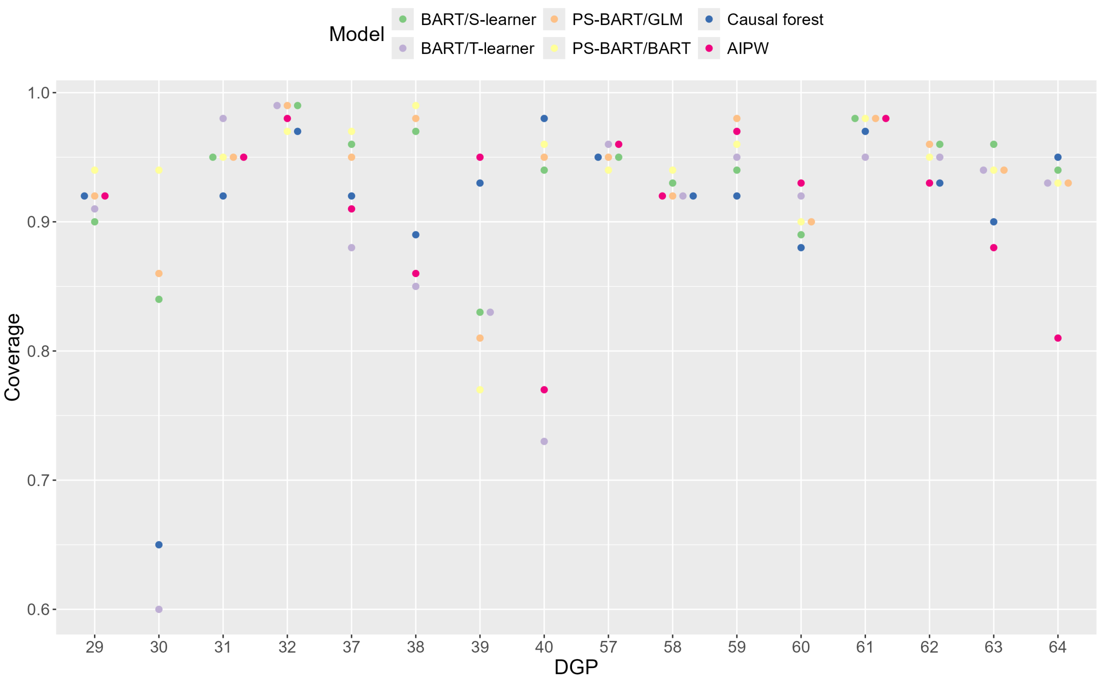


Figure 6.2: Coverage by model and DGP (AIPW's outlier in DGP 30 excluded).

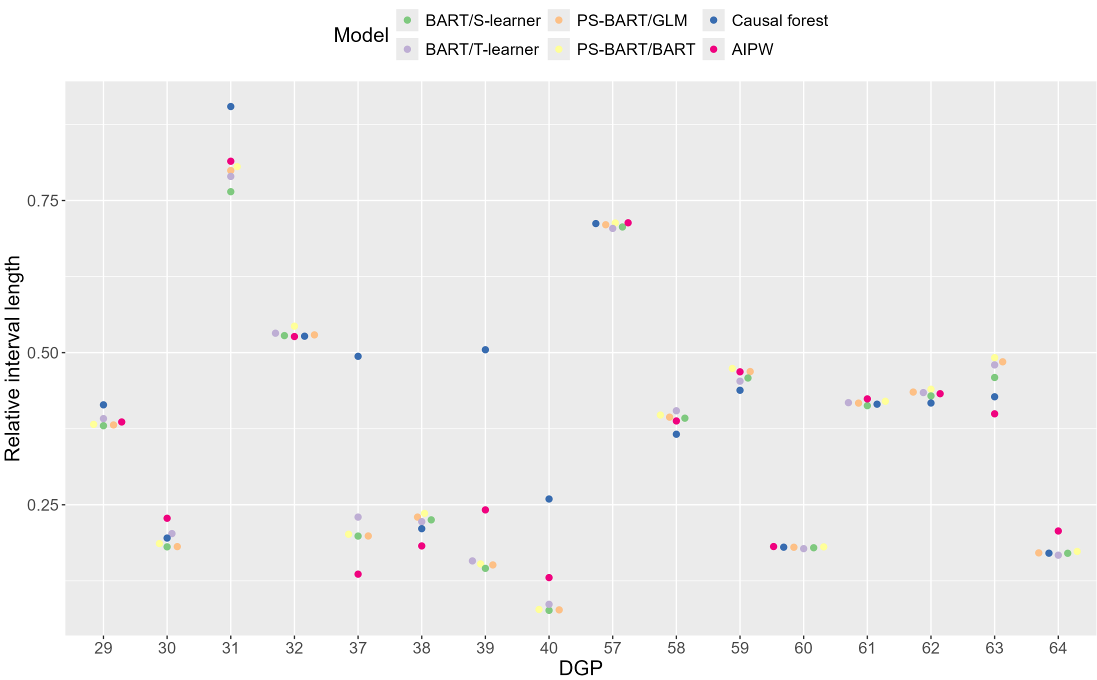


Figure 6.3: Relative interval length by model and DGP.

BART S-learners

Across all DGPs, the RMSE of the three S-learners variants (S-learner, PS-GLM, PS-BART) was almost always very similar. Therefore, they are discussed together. The biggest difference between the three occurred at DGP 30, where the RMSE of PS-BART (0.049) was substantially lower than that of the S-learner and PS-GLM (0.058 each).

One of the three S-learners placed first in half of the DGPs (8 of 16). Independently, the S-learner was the best model in 3 DGPs (31, 38, 61) and PS-BART in 1 (30). In the remaining 4 DGPs, first place was shared: all three S-learner variants tied at DGPs 40 and 62, the S-learner tied with the T-learner at DGP 32, and with AIPW at DGP 39.

Even when outperformed by other models, the margin was typically small, e.g. for DGP 64, the causal forest outperformed the best S-learner by only 0.002. The biggest difference between the worst S-learner and the best overall model is 0.014 between PS-BART and the T-learner at DGP 63. None of the S-Learners ranked last in any DGP.

When comparing the three S-learner variants only to each other, the S-learner was the sole best performer in 7 DGPs, PS-BART in 3, and PS-GLM never had the lowest RMSE on its own. There were several ties: all three models tied at DGPs 40 and 62, PS-GLM and PS-BART tied at DGP 60, and the S-learner and PS-GLM tied at DGPs 32, 59, and 64.

The pattern of small differences also extends to the coverage and interval length, which were similarly close across all DGPs. PS-BART had on average slightly higher coverage than the other two variants, consistent with its comparatively wider confidence intervals. All three BART S-learners reached a coverage rate of 0.90 or above for all but three DGPs. As discussed in Section **??**, DGP 30 and DGP 39 stand out with notably poor coverage. Coverage was at 0.95 or above in 8 cases for the S-learner, 9 cases for PS-GLM and 8 cases for PS-BART.

For most DGPs, the S-learner had the narrowest intervals among all BART models, closely followed by PS-GLM, while PS-BART's intervals were typically slightly wider.

T-learner

In most DGPs, the RMSE of the T-learner was close to that of the three S-learners. Most of the time it was slightly higher than all S-learners (e.g. in DGP 29, 58 and 61) or ranked somewhere between them (e.g. in DGP 32 or 39). Larger gaps to the S-learners appeared in DGPs 30, 37, 38, 40 and 62, with DGP30 showing the biggest difference of 0.044 to the S-learner and PS-GLM.

The T-learner outperformed all three S-learners in only four cases (DGPs 57, 59, 60, 63), in three of them (all except 60) it had the lowest RMSE of all models. The differences, however, were very small, with 0.005 at DGP 57 being the largest difference to the S-learner, which placed second. The T-learner had a lower RMSE than the causal forest in 10 DGPs and a lower RMSE than AIPW in 9 DGPs.

In 11 DGPs, the T-learner reached a coverage rate of 0.90 or above, in 6 DGPs above 0.95. Comparing the coverage rates to the S-learners, reveals a similar pattern as the RMSE. For DGPs, where differences between T-learner and S-learners were small, coverage rates were also similar. Coverage was substantially lower than that of the S-learners at DGPs 30 (0.60), 37 (0.88), 38 (0.85) and 40 (0.73). Those are DGPs where noticeable differences in RMSE were present. Coverage at DGP 39 was also relatively low (0.83), however, here all S-learners showed similarly low or even lower coverage.

In most DGPs, the interval lengths of the T-learner were slightly wider than those of the S-learners, but the differences were usually small. The largest difference was 0.031 at DGP 37 with 0.230 compared to 0.199 for the S-learner and PS-GLM. There were few exceptions where the T-learner had narrower intervals (e.g. DGP 59, DGP 60), but differences were marginal.

Causal forest

The causal forest had the highest RMSE in 9 cases, ranked 5th in 2 cases and 4th in 2 cases. In some cases (such as DGPs 29, 32, 60), there were no large differences to the model with the lowest RMSE (e.g. only 0.008 in DGP 32). However, in some DGPs, the RMSE of the causal forest was much higher than that of the best performing model. The notably largest gaps were in DGP 37, a difference of 0.110 to AIPW and in DGP 39, a difference of 0.092 to AIPW and the S-learner. When the causal forest did not rank last, it was usually still worse than the S-learners and only outperformed the T-Learner (in DGP 38) or both the T-learner and AIPW (in DGPs 30 and 62). There were two DGPs where the causal forest achieved the lowest RMSE out of all models (58 and 64) and one (63) where it ranked second and was outperformed by the T-learner by only 0.003. In all three cases, the difference to the next best model (in each case the S-learner) was very small (0.003 for DGP 58, 0.002 for DGP 63, and 0.002 for DGP 64).

The coverage of the causal forest was at 0.90 or above for 13 DGPs, in five of those at 0.95 or above, and slightly below 0.90 for DGP 38 (0.89) and DGP 60 (0.88), making its coverage the most stable among all models. The only outlier was DGP 30 (0.65), where the causal forest had a relatively high RMSE and most models had low coverage.

Confidence interval length of the causal forest was often comparable to that of the other models (e.g. DGP 32, 60) and in some cases smaller than that of the BART models (e.g. DGP 58, 59, 62, 63) There were, however, some DGPs where its intervals were much wider, the largest differences were DGPs 37, 39 and 40, where its intervals were double or almost double as long as those of the next-widest model (0.494 vs. 0.230 for the T-learner in DGP 37, 0.505 vs. 0.242 for AIPW at DGP 39 and 0.259 vs. 0.130 for AIPW in DGP 40).

Augmented Inverse Probability Weighting

AIPW had the highest RMSE in 4 DGPs and the second highest RMSE in 4 DGPs, often the causal forest was the only model doing worse. Differences to the best performing models were often moderate (e.g. 0.022 in DGP 62). Apart from DGP 30, the biggest difference to the best performing model was in DGP 64 (0.041 to the causal forest). It had the lowest RMSE in four DGPs (29, 37, 39, 60), however, in these DGPs all or most other models are very close.

The RMSE of AIPW at DGP 30 (1.196) stands out as an extreme outlier. The estimate was so far off that it resulted in a coverage of 0.00, which is by far the worst performance across the whole evaluation. Even though almost all other models, especially the T-learner and the causal forest, also had lower coverage rates for this DGP than on average, none of them came close to the failure of AIPW here. For the other DGPs, coverage was at 0.90 or above for 11 DGPs. Coverage rates were especially low at 40 (0.77) and 64 (0.81).

Interval length did not deviate much from the BART models on average, but in some DGPs intervals were notably shorter, while coverage was lower. This was the case in DGP 37, where interval length was 0.136 compared to the next shortest intervals (0.199 for the S-learner and PS-GLM), but coverage was lower (0.91) compared to 0.96 and 0.95. A similar pattern appeared in DGP 38 (AIPW: 0.182 vs. S-learner: 0.225; coverage 0.86 vs. 0.97) and DGP 63 (AIPW: 0.399 vs. S-learner: 0.459; coverage 0.88 vs. 0.96).

DGP 30 as an extreme outlier partly biased aggregated results. When excluding it from the calculation of the average metrics, would reduce the RMSE of AIPW from 0.171 to 0.102, while that of the causal forest would rise slightly from 0.117 to 0.119, making it the worst performing model instead. At the same time, AIPW's coverage would increase from 0.858 to 0.915, causing it to have a higher coverage than the T-learner (0.913).

6.1.3 Data Characteristic Measures

Relative and absolute number of observations in the smaller treatment group and with poor overlap can be found in Table 6.2. The numbers are averaged over 5 datasets.

Table 6.2: Overlap and balance measures by DGP

DGP	N No Overlap	No Overlap %	Treatment %	Smaller Group	Size Smaller Group
29	0.2	0.0%	30.6%	Treatment	1754
30	0.2	0.0%	68.4%	Control	1814
31	0	0.0%	60.0%	Control	2295
32	0	0.0%	45.8%	Treatment	2627
37	10.4	2.1%	17.8%	Treatment	89
38	0	0.0%	14.4%	Treatment	72
39	83.4	16.7%	32.3%	Treatment	162
40	15	3.0%	75.2%	Control	124
57	0	0.0%	37.4%	Treatment	243
58	0	0.0%	69.1%	Control	201
59	0	0.0%	52.2%	Control	310
60	1.2	0.2%	70.2%	Control	193
61	0	0.0%	65.8%	Control	222
62	0	0.0%	72.1%	Control	181
63	0.2	0.0%	51.0%	Control	318
64	0	0.0%	55.4%	Control	289

Balance

DGPs 37 and 38 have the highest levels of imbalance (17.8 % and 12.4% of observations in the treatment group), which resulted in a smaller treatment group with 89 and 62 observations. DGP 40 was imbalanced as well (76.2 % in the treatment group) but in the opposite direction, with a control group of 119 observations. In all other DGPs, the smaller group never fell below 28 %.

Across the three imbalanced DGPs, the T-learner had the lowest coverage of all models (0.88 in DGP 37, 0.85 in DGP 38 and 0.73 in DGP 40) and a moderately higher RMSE than the other BART variants. AIPW's coverage was also reduced relative to its performance in other DGPs (0.91 in DGP 37, 0.86 in DGP 38 and 0.77 in DGP 40). The causal forest had higher RMSE values in DGP 37 (0.149) and DGP 40 (0.069), but coverage stayed comparatively high (0.92 in DGP 37, 0.89 in DGP 38 and 0.98 in DGP 40). Coverage of the three S-learners stayed high across all three DGPs, ranging from 0.94 to 0.99, and RMSE values were lower than those of the T-learner and the causal forest.

Overlap

The Hill & Su (2013) method identified three DGPs where observations with poor overlap made up more than 1 %: DGP 39 (83.4 observations on average, 16.7 %), DGP 40 (15 observations, 3 %), and DGP 37 (10.4 observations, 2.1 %). DGP 39 is the only DGP where the observations without overlap make up a substantial proportion of the dataset.

In DGP 39, all BART models and AIPW had RMSE values close to each other, ranging from 0.053 to 0.058. AIPW's coverage was high at 0.95, but all four BART models showed reduced coverage (0.83 for the S-learner and T-learner, 0.81 for PS-GLM, and 0.77 for PS-BART). The causal forest showed a higher RMSE (0.145), but its coverage remained high (0.93).

6.2 Hyperparameter Tuning

The four DGPs that were chosen for hyperparameter tuning were DGPs 30, 40, 60 and 63. In DGP 30 and 40, the three S-learners achieved lower RMSE values than the other methods. In DGPs 60 and 63, the S-Learners were outperformed by other methods (AIPW and the T-learner in DGP 60 and the T-learner and causal forest in DGP 63).

As described in Section 5.9, the default performance reported in this section was recalculated using only the same 20 datasets used for tuning, and may therefore differ slightly from the results in Tables .1 to .4.

6.2.1 General Results

Overall, differences in performance between hyperparameter specifications were rather small. In many cases, the difference between the worst and the best performing specifications were smaller than the differences between methods.

For all BART models, the differences in RMSE between the best and the worst hyperparameter combinations was below 0.01 for DGPs 40 and below 0.05 for DGP 60. For DGP 30, only the T-learner exceeded 0.1 with a difference of 0.133 between the best $(k = 1, m = 200, \nu = 10, q = 0.75)$ and worst combination $(k = 5, m = 200, \nu = 10, q = 0.75)$, which is the largest difference among all BART models across all the included DGPs. The differences for DGP 63 were above 0.1 for all BART models except the T-learner, with PS-GLM having the highest difference of 0.114 between the best $(k = 1, m = 75, \nu = 3, q = 0.90)$ and worst combination $(k = 5, m = 200, \nu = 10, q = 0.75)$. For the causal forest, the differences in the RMSE between the best and the worst hyperparameter combination were also rather small. Only for DGP 30 it exceeded 0.1 at 0.189, between the best combination with RMSE $= 0.096$ ($min.node.size = 20, sample.fraction = 0.50, mtry = 26$) and the worst with RMSE $= 0.285$, which was shared by three combinations with $min.node.size \in 5, 10, 20$, all with $sample.fraction = 0.30$ and $mtry = 9$.

Given these relatively small performance differences, discussing all hyperparameter combinations in detail would provide limited insight. Therefore, only general patterns will be discussed. The detailed results for all hyperparameter specifications can be found in Tables .12 to .31.

Regarding the best hyperparameters combinations, results were relatively consistent across the BART learners. For DGPs 30, 60 and 63, lower values of k generally performed better than higher values, with RMSE consistently increasing from 1 to 5. For DGPs 60 and 63, these lower k values tended to work best in combination with a smaller numbers of trees, where 75 trees often outperformed 100 trees and 100 trees outperformed 200 trees. On the other hand, DGP 30 appeared to benefit from a larger number of trees. DGP 40 differed from the other settings, as higher values of k produced better results and a larger number of trees was better for the S-learner and PS-BART. For the variance-related hyperparameters ν and q no clear patterns emerged across DGPs and learners.

For the causal forest, a sample fraction of 0.5 generally yielded better results than a sample fraction of 0.3, with the exception of DGP 63. No consistent pattern emerged for *min.node.size*. For *mtry*, higher values (up to $\min(\sqrt{n_{cov}} + 20,\, n_{cov})$) led to better results for DGPs 30 and 40, whereas lower values (starting at $n_{cov}/3$) were preferred for DGP 60 and 63.

Coverage generally improved as RMSE decreased. In DGPs 30, 60, and 63, worse-performing combinations (e.g. higher k for the BART models or lower *mtry* for the causal forest) showed both higher RMSE and lower coverage, in some cases dropping to 0.00. DGP 40 was an exception, where coverage remained consistently high (0.90–1.00) across nearly all combinations. Interval length varied much less than RMSE or coverage across combinations for most models, with the causal forest's *mtry* parameter being the main exception.

For AIPW, the comparison is between the default logistic regression propensity model and the alternative BART-based model, instead of a hyperparameter grid. Using BART for the propensity score improved RMSE in DGP 30 and DGP 40. In DGP 30, the change was substantial, by 0.842 from 1.191 to 0.349, in DGP 40 it was smaller, by 0.006 from 0.043 to 0.037. In DGP 60, the RMSE stayed the same and in DGP 63 it increased slightly by 0.001. Coverage followed a similar pattern, improving by 0.10 in DGP 40, remaining unchanged in DGP 30 and DGP 60, and decreasing by 0.10 in DGP 63. Interval length decreased for all

DGPs except DGP 30.

6.2.2 Improvements by DGP

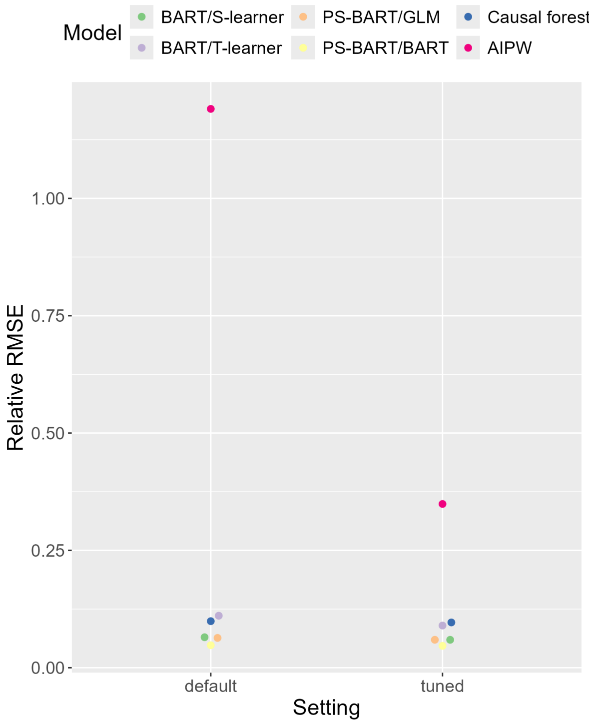
In addition to evaluating which hyperparameters yielded the best results overall, this section evaluates to what extent tuning improved model performance at each DGPs and how it changed the relative performance of the models. To do so, the best performing hyperparameter specification was compared with the default setting, Since many of the improvements were rather small, only the most notable changes are discussed. Figures 6.4a to 6.4d shows the RMSE differences between the default and tuned specifications. A more detailed overview can be found in Tables .8 to .11.

For DGP 30, hyperparameter tuning had little impact on the BART models and the causal forest. The most substantial improvement happened for AIPW, where estimating the propensity score with BART reduced the RMSE from 1.191 to 0.349. This is the largest improvement across all methods and DGPs. However, AIPW still was the worst performing method and coverage remained at 0.00. The ranking of the methods changed for the T-learner and the causal forest, with the T-learner slightly outperforming the causal forest by 0.006 when tuned, as well as for the S-learner and PS-GLM, which swapped places by a narrow margin of 0.001. Coverage increased by 0.05 for PS-GLM and the causal forest and by 0.10 for the S-learner and T-learner each.

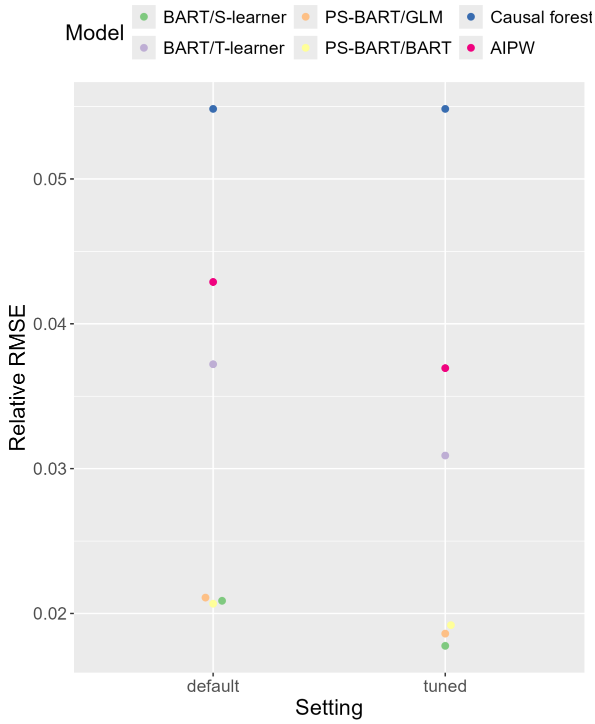
For DGP 40, hyperparameter tuning had virtually no impact on model performance. The largest decrease in RMSE could be observed at 0.006 for AIPW and the T-learner. For the causal forest, the default parameters actually were the best specification. Even though the S-learner only improved its performance by 0.003, it became the sole best performing model. In the default setting, it had shared the first place with PS-GLM and PS-BART. Coverage of the T-learner increased by 0.05 and of AIPW by 0.10.

For DGP 60, tuning again only resulted in small performance improvements. The largest RMSE reduction was 0.005 for PS-GLM, followed by S-learner and causal forest with 0.003. These improvements resulted in both PS-GLM and the S-learner surpassing PS-BART.Coverage increased by 0.05 for all four BART variants.

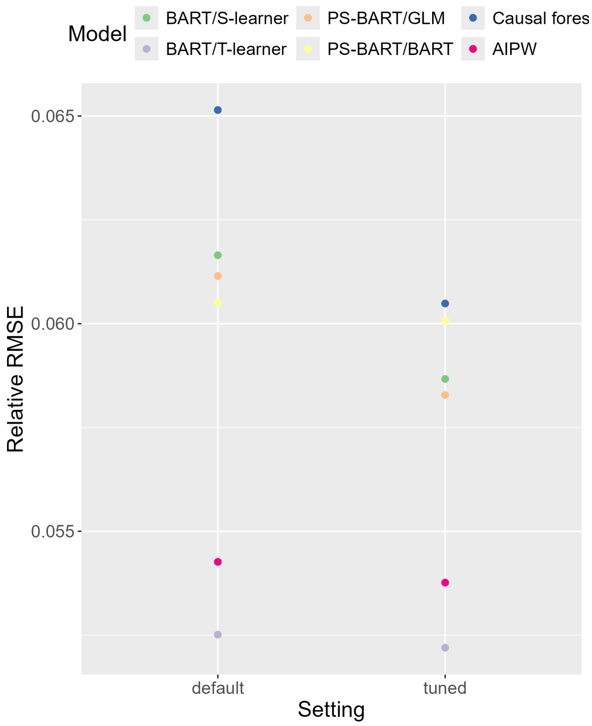
For DGP 63, the causal forest had the largest RMSE reduction of 0.019, followed by PS-GLM with 0.014 and the S-learner with 0.013. For AIPW, RMSE increased marginally by 0.001. The S-learner surpassed the T-learner and became the best performing method. Coverage increased by 0.05 for the S-learner, PS-GLM, and PS-BART, but decreased by 0.05 for the causal forest and by 0.10 for AIPW.



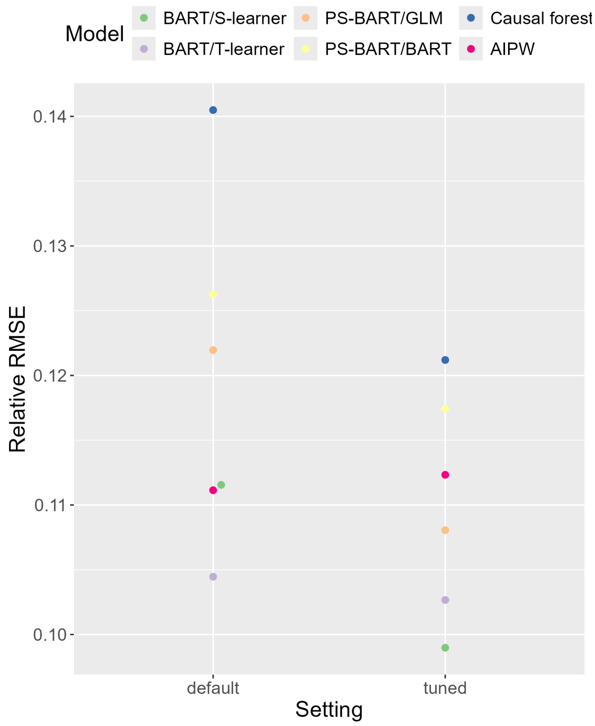
(a) DGP 30



(b) DGP 40



(c) DGP 60



(d) DGP 63

Figure 6.4: RMSE comparison of default and tuned model specifications

7 Discussion

7.1 Summary of Key Findings

The empirical results provide answers to the three research questions asked at the beginning of this thesis.

Among the BART models, the S-learner achieved the lowest average RMSE overall and had a higher coverage than the T-learner. In the few DGPs, where the T-learner outperformed the S-learner, differences were marginal. Compared to the basic S-learner, including estimated propensity scores improved performance only slightly or not at all in most settings. However, PS-BART achieved a slightly higher average coverage and yielded a notably better result than all other models in DGP 30.

Compared to the benchmark methods, the BART models were more consistently accurate. Even in DGPs where either AIPW or the causal forest slightly outperformed the BART models, the differences in RMSE were generally small.

Dataset characteristics played a role in model performance. Especially the coverage of the T-learner and AIPW was affected by imbalance in the datasets, while poor overlap affected the coverage of all BART models.

Hyperparameter tuning had only limited overall impact on model performance. RMSE differences between hyperparameter combinations were generally small. Tuning changed the relative ranking of the models, typically among models that were already close in the default setting.

7.2 Interpretation of the results

7.2.1 BART models

The S-learner clearly outperforming the T-learner in terms of RMSE and coverage in most settings contrasts with previous literature which suggests that both meta-learners have their strengths, with no single learner consistently dominating across studies (Künzel et al. (2019); see also Section 4.6). A more varied result with the T-learner outperforming the S-learner more clearly in some settings would therefore have been expected. For example, T-learners are expected to have an advantage when response surfaces of the treatment and control groups differ substantially (Caron et al., 2022). Since the data generating processes of the evaluated datasets are unknown, it is not possible to determine whether this characteristic was absent. Alternatively, this theoretical advantage of the T-learner may be true for other base learners, but become less pronounced when BART is used. The highly flexible nature of BART may allow an S-learner to model complex and heterogeneous response surfaces for both treatment groups within a single model and reduce the advantage of fitting separate models. Overall, the results suggest that the BART S-learner is a suitable default choice, as it achieved the best overall predictive accuracy while rarely performing worse than the T-learner.

Another unexpected finding was that PS-GLM and PS-BART did not perform better than the S-learner. The RMSE values of the three models were close for most DGPs, in fact, the S-learner was the model with the lowest average RMSE. This contradicts Hahn et al. (2020), who argued that including propensity scores makes the model more robust against regularization-induced confounding and supported this theoretical claim with empirical findings, a result also found by McConnell & Lindner (2019). While those studies did not compare different methods for propensity score estimation, it might additionally have been expected that BART would outperform a simple logistic regression as the propensity score model, given its greater flexibility. However, no substantial differences emerged between PS-GLM and PS-BART either. One notable exception was DGP 30, where PS-BART was the only model with coverage close to the nominal 0.95. This suggests that this DGP may have had a highly non-linear treatment assignment mechanism, which only the BART-based propensity score model was able to capture. In settings where complex treatment assignment is expected, including propensity scores estimated by BART may therefore provide an advantage. However, since it is unknown whether this characteristic was less pronounced in the other DGPs, it cannot be confirmed that this is specifically why propensity scores provided little benefit elsewhere. When treatment assignment is expected to be closer to linear, the choice between these BART models appears to matter far less.

Interval length differed only marginally among the four BART models. PS-BART had the longest intervals, followed by the T-learner, PS-GLM, and the S-learner with the shortest intervals. These differences likely reflect how much uncertainty the different models contain. The T-learner's wider intervals are likely caused by fitting two separate models, which introduces additional estimation uncertainty compared to the single model of the S-learner. Including propensity scores adds another level of uncertainty, since the propensity scores are estimates rather than a fixed covariate. A possible explanation why PS-BART's intervals are wider than those of PS-GLM is that PS-BART's propensity score is itself a posterior distribution and may introduce this additional estimation uncertainty into the outcome model.

7.2.2 Benchmark models

The causal forest often had a higher RMSE than the BART models, which is in line with previous literature (Caron et al., 2022; Hahn et al., 2020; Miratrix et al., 2025). However, as a tree-based method, it might be expected to share BART's flexibility in modeling complex, non-linear relationships. A possible explanation why it still performed worse is the honesty principle which splits the data into subsamples for tree construction and treatment effect estimation. While this approach is intended to reduce bias, it comes at the cost of using less data which may reduce estimation accuracy. In contrast, the Bayesian regularization of BART shrinks individual trees toward zero without requiring the sample to be split. Additionally, causal forests are optimized to identify treatment effect heterogeneity. If the true treatment effects in the ACIC DGPs were mostly homogeneous, this advantage might have been diminished.

Differences in coverage between the causal forest and the BART models were less pronounced. This is likely caused by the comparatively wide confidence intervals, which were the widest among all methods. While the estimates of the causal forest were less precise, it compensated for this shortcoming by providing more conservative confidence intervals.

In line with previous literature (Kim et al., 2023; McConnell & Lindner, 2019), AIPW was on average outperformed by the BART models. AIPW's ranking as the model with the highest average RMSE and lowest average coverage was partly driven by its extreme result at DGP 30. When this DGP is excluded, AIPW had a lower average RMSE than the causal forest and a higher coverage than the T-learner. While it was still outperformed by the BART S-learners on average, there were several DGPs where it yielded comparable or even better results. This suggests that AIPW's weakness is not a general lack of robustness, instead it can fail under specific circumstances. AIPW relies on the correct specification of either the treatment or the outcome model. Since both were modeled using a logistic regression in the default specification, its failures likely occurred in DGPs where one or both mechanisms were highly non-linear. In DGP 30, during the tuning step, AIPW's RMSE improved substantially when its treatment assignment mechanism was estimated by a more flexible BART model. However, much smaller improvements were seen in DGP 40, while DGP 60 remained the same and in DGP 63 the RMSE even increased. Furthermore, coverage in DGP 30 remained unchanged despite the large reduction in RMSE. This suggests that a more flexible propensity model can substantially improve AIPW's performance in some settings, likely where the treatment assignment mechanism is complex. On the other hand, it did not perform better than logistic regression, or even performed worse, in some cases. Therefore, a more flexible propensity score model does not seem to offer a consistent improvement across all settings, much like a more flexible hyperparameter specification does not for the BART learners.

Overall, no single model outperformed the others across all DGPs. Even the causal forest and AIPW achieved the best results in some DGPs. However, when they did, the margin to the BART models was usually small. On the other hand, when they were outperformed, the difference was usually larger. This suggests that in general BART models offer stronger accuracy and more consistent reliability across a range of data generating processes and are therefore preferable as a default choice among the evaluated methods.

7.2.3 Data Characteristics

Apart from these general results, the data chracateristic measures allowed for insights into how the performance of the different models was affected by imbalance and poor overlap.

The performance differences in the imbalanced datasets can likely be explained by the way the models are fit. The T-learner builds two separate models, one on the data of the treatment group, one on the data of the control group. When one of the groups has very few observations, variance increases, which lowers coverage rates. The S-learners fit one model with all the data, which avoids this issue. For AIPW, a small group can lead to extreme propensity scores which produce unstable weights, leading to higher variance (Cole & Hernán, 2008). The causal forest uses both groups jointly for splitting but estimates are made per leaf, so imbalance still destabilizes estimates. While this raises RMSE, the causal forest reflects this uncertainty in wider intervals, which is likely why it did not show the same drop in coverage as the T-learner and AIPW.

In contrast to the imbalanced DGPs, where the S-learner variants remained largely unaffected, poor overlap appears to affect all BART variants similarly. A possible explanation is that all BART models estimate treatment effects by imputing the potential missing outcomes for each observation. In a region with poor overlap, the model has to extrapolate (Hill & Su, 2013). The causal forest does not impute missing outcomes but estimates treatment effects within each leaf. In a region with poor overlap, a leaf may contain only very few observations from one of the groups, which can result in less precise estimates. AIPW appeared largely unaffected by poor overlap, which may reflect that its doubly robust property can compensate for an outcome model which is not correctly specified in every region (Robins et al., 1994).

DGP 37 and 40 also showed regions of missing overlap, but at a much lower level than DGP 39. Since both DGPs were also identified as imbalanced, it is difficult to say whether the bad performance of some models was because of poor overlap, imbalance, or both. As the missing overlap in these two DGPs was much smaller than in DGP 39, it is possible that most of the performance differences were driven mainly by imbalance rather than overlap.

In general, the findings for both data characteristic measures should be interpreted with caution, especially for overlap, since only one DGP with poor overlap was identified, so it was not possible to observe whether the pattern would replicate over multiple DGPs.

7.2.4 Hyperparameter tuning

Hyperparameter tuning had only a limited impact on results overall. The differences between the best and worst performing specifications were mostly small for all BART models and the causal forest. As discussed in Section 7.2.2, AIPW's performance in DGP 30 improved drastically when its treatment assignment mechanism was changed from a logistic regression to a BART model. However, this improvement was not consistent across DGPs. While this is not the same as hyperparameter tuning, it suggests that AIPW's performance can depend considerably more on the flexibility of its nuisance models than BART's or the causal forest's performance depends on their hyperparameters, at least in settings where the underlying mechanisms are complex.

For the BART models, $k = 1$ outperformed the default ($k = 2$) in all DGPs except DGP 40, suggesting slightly less shrinkage is favored. The pattern for the number of trees m was less consistent: a larger m tended to perform better in DGPs 30 and 40, while a smaller m performed better in DGPs 60 and 63. For the variance-related hyperparameters ν and q, no clear pattern emerged across DGPs or learners, suggesting a comparatively weaker influence on performance than k or m. Given that these patterns were inconsistent across DGPs and the resulting improvements were small, they do not provide a strong basis for recommending different default values. Especially for m the recommended value 75 should remain, since increasing the number of trees increases computational cost without guaranteeing a gain in performance.

For the causal forest, the default $mtry$ value performed well in DGPs 30 and 40, but a lower value performed better in DGPs 60 and 63. In DGP 30, which likely had a complex treatment assignment mechanism (see Section 7.2.2), a higher number of $mtry$ had the biggest impact. This suggests that a higher number of covariates considered at each split might allow the causal forest to better capture complex mechanisms. The $sample.fraction$

default value of 0.5 seemed to work well except for DGP 63 and for *min.node.size* no consistent pattern emerged. Based on these findings, keeping the default values seems to be reasonable with the exception of *mtry*, which may be worth tuning.

Hyperparameter tuning also had only a limited influence on the ranking of the methods. Where the ranking changed,it was usually between methods that were already close to each other under default settings. When differences between methods were larger under default settings, rankings did not change through tuning. This suggests that the choice of method had a considerably larger impact on performance than the choice of hyperparameters within a given method. This contradicts Machlanski et al. (2023), who found hyperparameter tuning to have a greater impact on causal effect estimation performance than the choice of causal estimator or base learner.

Overall, based on the findings from this thesis, it appears to be a sensible choice to use the default hyperparameters for BART models and the causal forest. The choice of the method itself appears to matter more than the choice of hyperparameters within a method.

7.3 Limitations

A methodological strength of this study is that the data was not simulated within the study but taken from another source. Not knowing the underlying data generating processes minimized the risk of choosing or adapting datasets in a way that would favor particular methods. At the same time, this is one of the study's main limitations. Since there is no knowledge about what the covariates represent, how treatment assignment is determined and how covariates, treatment and outcome interact, it is not possible to draw conclusions about which data characteristics influence the performance differences observed between models. While balance was measured and overlap was estimated to draw some conclusions, they were not directly manipulated and other relevant characteristics remain completely unknown. This is particularly relevant for the overlap measure, since only one DGP with poor overlap was identified, and the method used to identify it relies on the posterior variance of BART, so some correlation with poor performance of the BART models was to be expected. The findings on overlap may therefore partly be a consequence of the measuring method itself rather than an independent confirmation of the underlying pattern. 16 different DGPs were used and their creators claimed that they cover a diverse range of scenarios, so it is reasonable to assume that the findings generalize to a wide range of data generating processes. Still, it can not be completely ruled out that the DGPs were still biased toward certain methods.

Another limitation of this thesis is that it only focused on the estimation of the average treatment effect. Since the creators of the evaluated datasets did not publish the true individual or conditional average treatment effects, it was not possible to assess how accurately the methods estimated treatment effect heterogeneity. This may be particularly relevant for the causal forest, which was specifically designed to detect heterogeneous treatment effect. In the evaluation its ATE estimates were less accurate compared to the BART models, but it might yield better estimates of heterogeneous treatment effects.

A further limitation concerns the scope of data conditions considered in this thesis. This thesis focused on low-dimensional datasets with binary treatment assignment and continuous outcomes. As a consequence, findings should only be generalized to other settings with consideration. The datasets from the ACIC Data Challenge 2019 contained high-dimensional datasets as well. Compared with low-dimensional datasets used in this thesis, which contained at most 31 covariates, it would be interesting to see how the evaluated methods perform in a high-dimensional setting. Berkessa et al. (2025) summarized which challenges can appear in high-dimensional data, e.g. spurious correlation, so correlations between unrelated covariates which occur purely as a result of high dimensionality, or endogeneity. Additionally, the datasets from the ACIC Data Challenge 2019 contained binary outcome variables. Treatment effects for binary outcomes are less straightforward to estimate and interpret, since they are usually expressed as odds ratios or risk differences (Hu et al., 2020). It remains unknown how the methods evaluated here would perform under high-dimensional datasets or with binary outcomes.

This thesis evaluated the performance of different BART meta-learners. Specifically, an S-Learner, two variants of the S-Learner that incorporated propensity scores as additional covariates, and a T-Learner were compared. However other meta-learning approaches exist, such as the X-Learner or the R-Learner. The X-learner has been shown to outperform the S- and T-learner in some settings (Künzel et al., 2019), and the R-learner has similarly demonstrated strong performance relative to existing methods (Nie & Wager, 2021). Furthermore, Hahn et al. (2020) introduced an extension of the standard BART model, the Bayesian Causal Forest (BCF). BCF models the baseline outcome function and treatment effects separately, which improves robustness in the presence of targeted selection and confounding. Compared with standard BART and propensity score-augmented BART models, BCF often produces less biased estimates and achieves higher coverage rates (e.g., Hahn et al., 2020; McConnell & Lindner, 2019). Neither of these alternative BART-based methods were included in the present evaluation.

Furthermore, hyperparameter tuning was performed on only four selected DGPs and 20 datasets each compared to the 100 in the default specification. Results can therefore only be interpreted regarding those DGPs, which may have different underlying mechanisms. Since the DGPs were chosen to represent the whole range of number of covariates, number of observations and different best-performing models in the default specification, some kind of generalizability should still be assumed.

More importantly, hyperparameter tuning in this thesis does not fully reflect how model selection is performed in real-world applications. The different hyperparameter specifications were evaluated by comparing the resulting ATE estimates with the known true ATE values. For this thesis, this was an appropriate approach as the objective was to evaluate whether hyperparameter tuning improves estimation accuracy or changes the relative performance among the models. In contrast, when researchers aim to select the best parameters for a model in practical applications, they usually rely on cross-validation. This approach is well established for predictive tasks, however, it is not directly applicable for causal inference because the true treatment effects are generally unknown. Hyperparameter tuning for causal models is therefore faced with the challenge of evaluating which specification achieves the most accurate estimates without observing the ground truth. There are different approaches to address this issue, such as comparing the predicted outcomes to the observed outcomes or using plug-in estimators as proxy performance measures. All of these methods have their own limitations. Machlanski et al. (2023) compared several methods and found that their performance differed between datasets, so currently no universally accepted tuning criterion exists.

7.4 Future research

Building on the discussed limitations, several opportunities for future research emerge.

First, future studies could systematically evaluate how model performance varies with different dataset characteristics. Synthetic datasets, in which characteristics such as overlap, treatment effect heterogeneity or the degree of non-linearity in the treatment assignment and outcome mechanisms are systematically varied, would allow more concrete findings regarding their effects on model performance.

Second, the scope of methods and data settings could be extended. Additional BART-based meta-learners, e.g. as the X-learner or R-learner, as well as the Bayesian Causal Forest could be compared to the models evaluated in this thesis. It would be especially interesting to see how the S-learner, which emerged as the most reliable model in this evaluation, compares to those variants. The evaluation could also extend to high-dimensional datasets and to binary outcomes. The `dbarts`, `bartCause`, `grf` and `AIPW` packages used in this thesis all allow for binary outcomes, which allows for easy implementation. Further extensions could include multi-categorical or continuous treatments.

Third, this thesis was limited to the estimation of average treatment effects. Future research should assess how well different methods estimate heterogeneous treatment effects. Especially the causal forest, which is designed to detect treatment effect heterogeneity, might have an advantage over the BART models here. A common evaluation metric for this purpose is the Precision in Estimation of Heterogeneous Effects (PEHE) (Hill, 2011).

Finally, future work could examine approaches for when the assumptions for causal inference are not fulfilled. This thesis already identified regions of poor overlap, in a next step those regions could be excluded from estimation. Crump et al. (2009) found that discarding observations with extreme propensity scores can improve accuracy, so assessing whether this also works for the evaluated methods would be a valuable direction for future research. Additionally, when the ignorability assumption does not hold, sensitivity analysis approaches (e.g. by VanderWeele & Ding (2017)) could explore how robust estimates are against unobserved confounders.

8 Conclusion

This thesis evaluated how the machine learning model Bayesian Additive Regression Trees performs when applied to causal effect estimation. Specifically, it aimed to answer three research questions:

1. How does the performance of different causal BART learners compare?
2. How do BART-based approaches compare to other causal effect estimation methods?
3. To what extent can hyperparameter tuning improve model performance, and can tuning alter which method performs best?

To answer these questions, four meta-learners that use BART as their base learner were compared: an S-learner, a T-learner, and two S-learners that included propensity scores as additional covariates (one estimated with a logistic regression, one with a BART model). They were compared against each other and against two benchmark causal effect estimation methods, a doubly robust augmented inverse probability weighting estimator and a causal forest. The evaluation used 16 synthetic and semi-synthetic data-generating processes from the ACIC 2019 Data Challenge, with 100 datasets each. The models were assessed in terms of RMSE, coverage, and interval length. The level of imbalance and the degree of overlap were computed as data characteristics. Hyperparameter tuning was performed on a subset of four DGPs for all models.

Among the BART learners, the S-learner performed best overall. The T-learner performed worse on average, particularly under imbalanced data. All BART models were negatively affected by poor overlap, but only one DGP with low overlap was identified, so this finding is limited. Including estimates of the propensity score improved results when the treatment assignment mechanism was complex, but it often did not improve results substantially otherwise. While the causal forest and AIPW sometimes achieved results close to or better than the BART learners, they were outperformed by BART overall.

Hyperparameter tuning showed only marginal changes, as the default hyperparameters of BART and the causal forest seemed to be well tuned already, and different specifications had little effect on performance. Tuning also rarely changed which method performed best.

This suggests that, within the range of methods and DGPs evaluated here, the choice of method matters more than the choice of hyperparameters.

This thesis contributes to the existing literature by providing a systematic comparison of causal BART meta-learners against each other and against established benchmark methods across a broad range of data-generating processes. For practitioners, the results suggest that the BART S-learner, used with its default hyperparameters, is a reasonable default choice for average treatment effect estimation.

This thesis also has some shortcomings, mostly not being able to draw direct conclusions about the relationship between data characteristics and model performance, and the limited scope of the evaluated settings. These shortcomings offer concrete possibilities for future research, such as extending the comparison to heterogeneous treatment effects, additional meta-learners, and high-dimensional or binary-outcome settings. Overall, this thesis provides evidence that BART is a reliable choice for average treatment effect estimation among the growing variety of causal machine learning methods.

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Boxplots of Model Performance

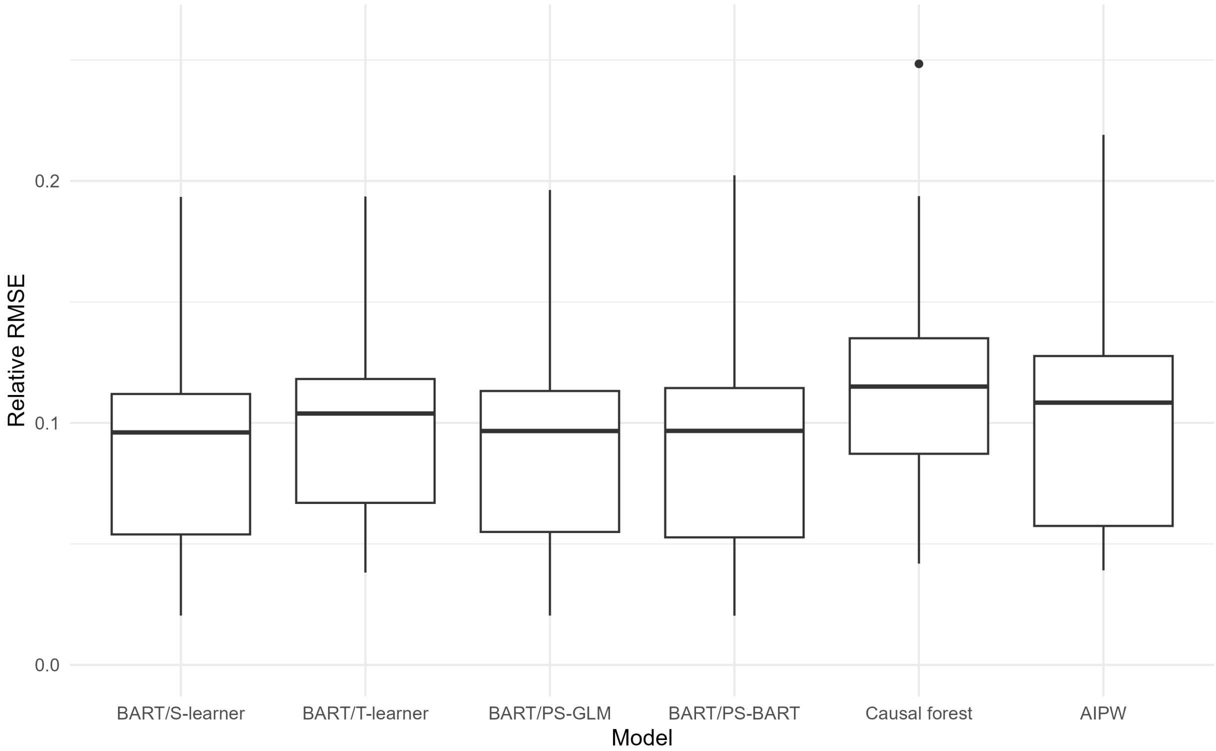

Figure .1: Distribution of relative RMSE across DGPs by model (AIPW's outlier in DGP 30 excluded)

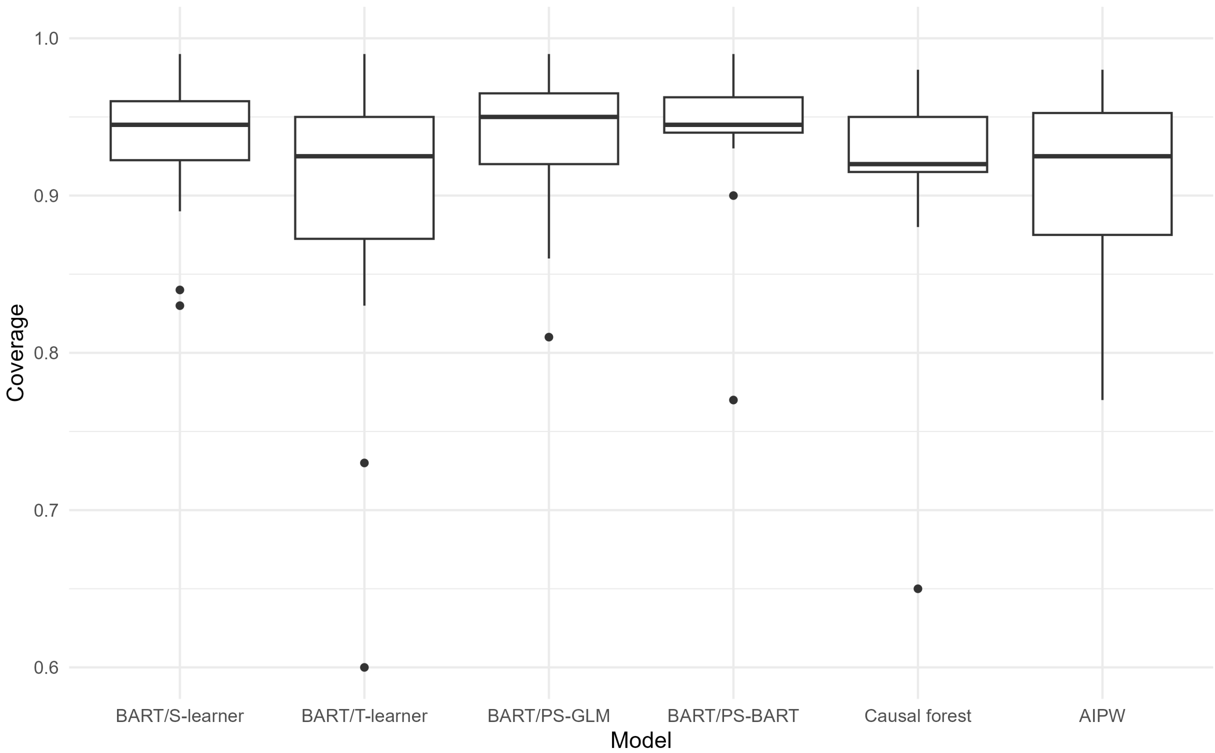

Figure .2: Distribution of coverage across DGPs by model (AIPW's outlier in DGP 30 excluded)

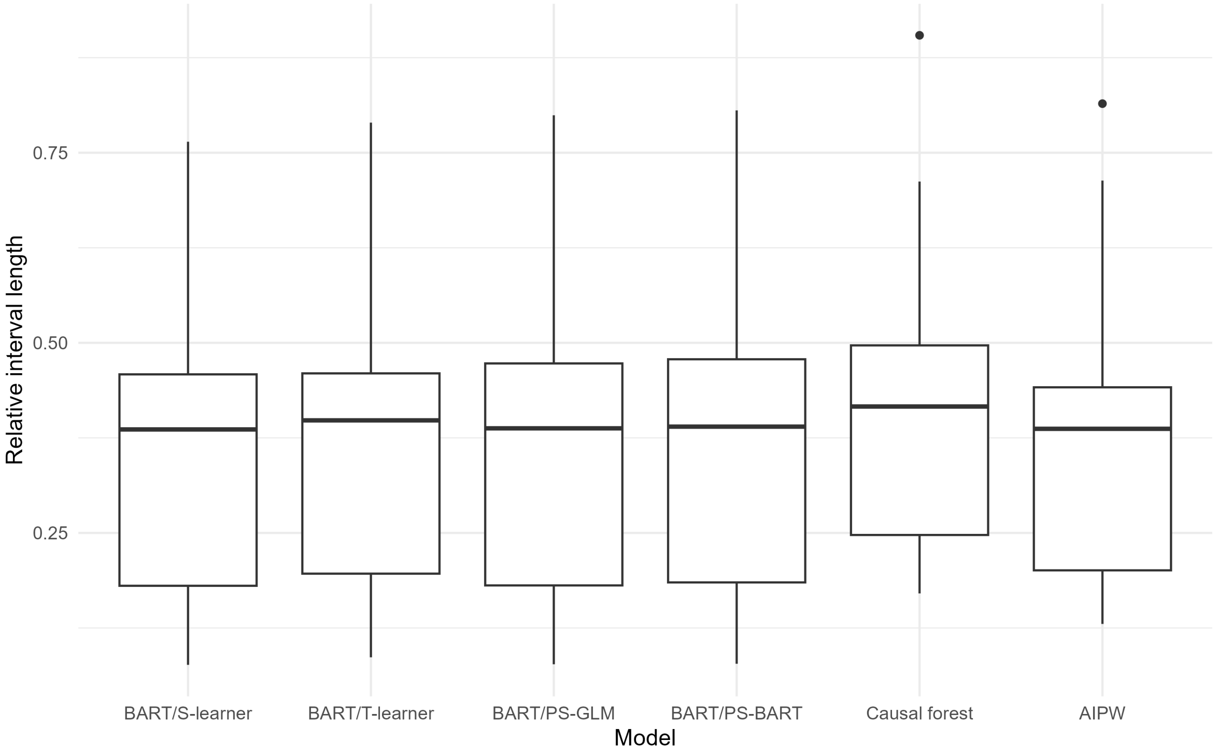

Figure .3: Distribution of relative interval length across DGPs by model

Aggregated Results by DGP and Model

Table .1: Relative RMSE, coverage and relative interval length by model and DGP

Model	DGP	RMSE	Coverage	Interval Length
BART/S-learner	DGP 29	0.109	0.90	0.380
BART/T-learner	DGP 29	0.116	0.91	0.392
BART/PS-GLM	DGP 29	0.109	0.92	0.381
BART/PS-BART	DGP 29	0.108	0.94	0.382
Causal forest	DGP 29	0.116	0.92	0.414
AIPW	DGP 29	0.107	0.92	0.386
BART/S-learner	DGP 30	0.058	0.84	0.181
BART/T-learner	DGP 30	0.102	0.60	0.203
BART/PS-GLM	DGP 30	0.058	0.86	0.181
BART/PS-BART	DGP 30	0.049	0.94	0.186
Causal forest	DGP 30	0.093	0.65	0.195
AIPW	DGP 30	1.196	0.00	0.228
BART/S-learner	DGP 31	0.193	0.95	0.764
BART/T-learner	DGP 31	0.194	0.98	0.790
BART/PS-GLM	DGP 31	0.196	0.95	0.799
BART/PS-BART	DGP 31	0.202	0.95	0.806
Causal forest	DGP 31	0.248	0.92	0.904
AIPW	DGP 31	0.219	0.95	0.815
BART/S-learner	DGP 32	0.124	0.99	0.528
BART/T-learner	DGP 32	0.124	0.99	0.532
BART/PS-GLM	DGP 32	0.124	0.99	0.529
BART/PS-BART	DGP 32	0.128	0.97	0.543
Causal forest	DGP 32	0.132	0.97	0.527
AIPW	DGP 32	0.127	0.98	0.526

Table .2: Relative RMSE, coverage and relative interval length by model and DGP

Model	DGP	RMSE	Coverage	Interval Length
BART/S-learner	DGP 37	0.052	0.96	0.199
BART/T-learner	DGP 37	0.070	0.88	0.230
BART/PS-GLM	DGP 37	0.052	0.95	0.199
BART/PS-BART	DGP 37	0.051	0.97	0.202
Causal forest	DGP 37	0.149	0.92	0.494
AIPW	DGP 37	0.039	0.91	0.136
BART/S-learner	DGP 38	0.056	0.97	0.225
BART/T-learner	DGP 38	0.074	0.85	0.222
BART/PS-GLM	DGP 38	0.057	0.98	0.230
BART/PS-BART	DGP 38	0.059	0.99	0.236
Causal forest	DGP 38	0.062	0.89	0.211
AIPW	DGP 38	0.059	0.86	0.182
BART/S-learner	DGP 39	0.053	0.83	0.145
BART/T-learner	DGP 39	0.057	0.83	0.158
BART/PS-GLM	DGP 39	0.056	0.81	0.151
BART/PS-BART	DGP 39	0.058	0.77	0.153
Causal forest	DGP 39	0.145	0.93	0.505
AIPW	DGP 39	0.053	0.95	0.242
BART/S-learner	DGP 40	0.020	0.94	0.077
BART/T-learner	DGP 40	0.038	0.73	0.086
BART/PS-GLM	DGP 40	0.020	0.95	0.077
BART/PS-BART	DGP 40	0.020	0.96	0.078
Causal forest	DGP 40	0.069	0.98	0.259
AIPW	DGP 40	0.051	0.77	0.130

Table .3: Relative RMSE, coverage and relative interval length by model and DGP

Model	DGP	RMSE	Coverage	Interval Length
BART/S-learner	DGP 57	0.168	0.95	0.706
BART/T-learner	DGP 57	0.163	0.96	0.704
BART/PS-GLM	DGP 57	0.170	0.95	0.710
BART/PS-BART	DGP 57	0.173	0.94	0.713
Causal forest	DGP 57	0.194	0.95	0.712
AIPW	DGP 57	0.174	0.96	0.713
BART/S-learner	DGP 58	0.107	0.93	0.392
BART/T-learner	DGP 58	0.111	0.92	0.404
BART/PS-GLM	DGP 58	0.109	0.92	0.394
BART/PS-BART	DGP 58	0.110	0.94	0.397
Causal forest	DGP 58	0.104	0.92	0.366
AIPW	DGP 58	0.110	0.92	0.388
BART/S-learner	DGP 59	0.109	0.94	0.458
BART/T-learner	DGP 59	0.106	0.95	0.453
BART/PS-GLM	DGP 59	0.109	0.98	0.469
BART/PS-BART	DGP 59	0.110	0.96	0.474
Causal forest	DGP 59	0.123	0.92	0.438
AIPW	DGP 59	0.115	0.97	0.468
BART/S-learner	DGP 60	0.054	0.89	0.179
BART/T-learner	DGP 60	0.049	0.92	0.178
BART/PS-GLM	DGP 60	0.053	0.90	0.180
BART/PS-BART	DGP 60	0.053	0.90	0.181
Causal forest	DGP 60	0.058	0.88	0.180
AIPW	DGP 60	0.048	0.93	0.181

Table .4: Relative RMSE, coverage and relative interval length by model and DGP

Model	DGP	RMSE	Coverage	Interval Length
BART/S-learner	DGP 61	0.087	0.98	0.413
BART/T-learner	DGP 61	0.100	0.95	0.418
BART/PS-GLM	DGP 61	0.088	0.98	0.417
BART/PS-BART	DGP 61	0.088	0.98	0.420
Causal forest	DGP 61	0.105	0.97	0.415
AIPW	DGP 61	0.094	0.98	0.424
BART/S-learner	DGP 62	0.105	0.96	0.429
BART/T-learner	DGP 62	0.124	0.95	0.434
BART/PS-GLM	DGP 62	0.105	0.96	0.435
BART/PS-BART	DGP 62	0.105	0.95	0.440
Causal forest	DGP 62	0.114	0.93	0.417
AIPW	DGP 62	0.127	0.93	0.433
BART/S-learner	DGP 63	0.120	0.96	0.459
BART/T-learner	DGP 63	0.115	0.94	0.480
BART/PS-GLM	DGP 63	0.125	0.94	0.485
BART/PS-BART	DGP 63	0.129	0.94	0.492
Causal forest	DGP 63	0.118	0.90	0.427
AIPW	DGP 63	0.130	0.88	0.399
BART/S-learner	DGP 64	0.044	0.94	0.170
BART/T-learner	DGP 64	0.046	0.93	0.167
BART/PS-GLM	DGP 64	0.044	0.93	0.171
BART/PS-BART	DGP 64	0.046	0.93	0.173
Causal forest	DGP 64	0.042	0.95	0.170
AIPW	DGP 64	0.083	0.81	0.207

Model Rank Summaries by Metric

Table .5: Ranking of models by RMSE across all 16 DGPs (rank 1 = lowest RMSE)

Model	1st	2nd	3rd	4th	5th	6th	Mean Rank
BART/S-learner	7	5	3	0	1	0	1.94
BART/PS-GLM	3	5	7	1	0	0	2.38
BART/PS-BART	3	3	2	5	3	0	3.12
BART/T-learner	4	2	0	3	5	2	3.56
AIPW	4	0	1	3	4	4	3.94
Causal forest	2	1	0	2	2	9	4.75

Table .6: Ranking of models by coverage across all 16 DGPs (rank 1 = highest coverage)

Model	1st	2nd	3rd	4th	5th	6th	Mean Rank
BART/PS-GLM	4	5	6	0	1	0	2.31
BART/PS-BART	6	3	4	0	1	2	2.56
BART/S-learner	4	4	4	1	2	1	2.75
AIPW	4	3	1	1	4	3	3.44
BART/T-learner	3	2	4	1	2	4	3.56
Causal forest	2	2	2	3	4	3	3.88

Table .7: Ranking of models by interval length across all 16 DGPs (rank 1 = shortest interval)

Model	1st	2nd	3rd	4th	5th	6th	Mean Rank
BART/S-learner	6	5	4	1	0	0	2.00
BART/PS-GLM	2	3	4	3	4	0	3.25
Causal forest	3	5	1	2	0	5	3.38
BART/T-learner	3	2	1	5	4	1	3.50
AIPW	4	1	1	2	5	3	3.75
BART/PS-BART	0	0	4	2	5	5	4.69

Performance Comparison of Default and Tuned Hyperparameters

Table .8: DGP 30: Performance comparison of default and tuned model specifications

Model	Setting	k	m	ν	q	min.node.size	sample.fraction	mtry	RMSE (rel.)	Coverage	CI length (rel.)
BART/S-learner	default	2	75	3	0.90	–	–	–	0.065	0.75	0.178
BART/S-learner	tuned	1	200	10	0.75	–	–	–	0.059	0.85	0.179
BART/T-learner	default	2	75	3	0.90	–	–	–	0.111	0.50	0.199
BART/T-learner	tuned	1	200	10	0.75	–	–	–	0.090	0.60	0.202
PS-BART/GLM	default	2	75	3	0.90	–	–	–	0.064	0.75	0.180
PS-BART/GLM	tuned	1	200	3	0.99	–	–	–	0.060	0.80	0.178
PS-BART/BART	default	2	75	3	0.90	–	–	–	0.048	0.95	0.184
PS-BART/BART	tuned	3	100	10	0.75	–	–	–	0.047	0.95	0.183
Causal forest	default	–	–	–	–	5	0.5	26	0.099	0.50	0.194
Causal forest	tuned	–	–	–	–	20	0.5	26	0.096	0.55	0.194
AIPW	Logistic regression	–	–	–	–	–	–	–	1.191	0.00	0.225
AIPW	BART	–	–	–	–	–	–	–	0.349	0.00	0.252

Table .9: DGP 40: Performance comparison of default and tuned model specifications

Model	Setting	k	m	ν	q	min.node.size	sample.fraction	mtry	RMSE (rel.)	Coverage	CI length (rel.)
BART/S-learner	default	2	75	3	0.90	–	–	–	0.021	0.95	0.078
BART/S-learner	tuned	5	100	3	0.99	–	–	–	0.018	0.95	0.075
BART/T-learner	default	2	75	3	0.90	–	–	–	0.037	0.85	0.087
BART/T-learner	tuned	5	75	3	0.99	–	–	–	0.031	0.90	0.112
PS-BART/GLM	default	2	75	3	0.90	–	–	–	0.021	0.95	0.078
PS-BART/GLM	tuned	5	100	3	0.99	–	–	–	0.019	0.95	0.076
PS-BART/BART	default	2	75	3	0.90	–	–	–	0.021	0.95	0.079
PS-BART/BART	tuned	5	200	10	0.75	–	–	–	0.019	0.95	0.080
Causal forest	default	–	–	–	–	5	0.5	22	0.055	1.00	0.228
Causal forest	tuned	–	–	–	–	5	0.5	22	0.055	1.00	0.228
AIPW	Logistic regression	–	–	–	–	–	–	–	0.043	0.80	0.126
AIPW	BART	–	–	–	–	–	–	–	0.037	0.90	0.115

Table .10: DGP 60: Performance comparison of default and tuned model specifications

Model	Setting	k	m	ν	q	min.node.size	sample.fraction	mtry	RMSE (rel.)	Coverage	CI length (rel.)
BART/S-learner	default	2	75	3	0.90	–	–	–	0.062	0.90	0.180
BART/S-learner	tuned	1	75	3	0.90	–	–	–	0.059	0.95	0.182
BART/T-learner	default	2	75	3	0.90	–	–	–	0.053	0.90	0.177
BART/T-learner	tuned	1	75	3	0.99	–	–	–	0.052	0.95	0.181
PS-BART/GLM	default	2	75	3	0.90	–	–	–	0.061	0.90	0.181
PS-BART/GLM	tuned	1	75	3	0.90	–	–	–	0.058	0.95	0.182
PS-BART/BART	default	2	75	3	0.90	–	–	–	0.061	0.90	0.181
PS-BART/BART	tuned	2	75	10	0.75	–	–	–	0.060	0.95	0.184
Causal forest	default	–	–	–	–	5	0.5	26	0.065	0.85	0.180
Causal forest	tuned	–	–	–	–	5	0.5	11	0.060	0.85	0.180
AIPW	Logistic regression	–	–	–	–	–	–	–	0.054	0.95	0.181
AIPW	BART	–	–	–	–	–	–	–	0.054	0.95	0.160

Table .11: DGP 63: Performance comparison of default and tuned model specifications

Model	Setting	k	m	ν	q	min.node.size	sample.fraction	mtry	RMSE (rel.)	Coverage	CI length (rel.)
BART/S-learner	default	2	75	3	0.90	–	–	–	0.112	0.95	0.462
BART/S-learner	tuned	1	75	10	0.75	–	–	–	0.099	1.00	0.467
BART/T-learner	default	2	75	3	0.90	–	–	–	0.104	0.95	0.482
BART/T-learner	tuned	3	75	3	0.99	–	–	–	0.103	0.95	0.459
PS-BART/GLM	default	2	75	3	0.90	–	–	–	0.122	0.95	0.486
PS-BART/GLM	tuned	1	75	3	0.90	–	–	–	0.108	1.00	0.498
PS-BART/BART	default	2	75	3	0.90	–	–	–	0.126	0.95	0.496
PS-BART/BART	tuned	1	75	3	0.99	–	–	–	0.117	1.00	0.525
Causal forest	default	–	–	–	–	5	0.5	26	0.140	0.85	0.431
Causal forest	tuned	–	–	–	–	50	0.3	10	0.121	0.80	0.367
AIPW	Logistic regression	–	–	–	–	–	–	–	0.111	0.95	0.401
AIPW	BART	–	–	–	–	–	–	–	0.112	0.85	0.346

Detailed Hyperparameter Tuning Results

Table .12: Hyperparameter tuning results for S-learner in DGP 30

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
1	200	10	0.75	0.059	0.85	0.179
1	200	3	0.90	0.060	0.85	0.179
1	200	3	0.99	0.060	0.85	0.179
1	100	3	0.99	0.061	0.80	0.179
1	100	10	0.75	0.062	0.80	0.179
2	200	3	0.90	0.062	0.80	0.180
1	100	3	0.90	0.062	0.75	0.178
2	200	3	0.99	0.062	0.75	0.178
2	75	10	0.75	0.063	0.85	0.179
2	100	10	0.75	0.063	0.80	0.179
2	100	3	0.99	0.063	0.75	0.178
2	200	10	0.75	0.063	0.75	0.180
1	75	3	0.90	0.063	0.75	0.178
2	100	3	0.90	0.064	0.75	0.179
1	75	10	0.75	0.064	0.75	0.177
1	75	3	0.99	0.064	0.75	0.179
3	100	3	0.90	0.064	0.80	0.179
3	75	3	0.90	0.064	0.75	0.179
2	75	3	0.90	0.064	0.75	0.179
3	100	10	0.75	0.065	0.75	0.179
3	100	3	0.99	0.065	0.75	0.178
3	200	3	0.90	0.065	0.75	0.179
3	75	3	0.99	0.065	0.75	0.179
2	75	3	0.99	0.066	0.75	0.178
3	75	10	0.75	0.066	0.75	0.180
3	200	3	0.99	0.066	0.75	0.178
3	200	10	0.75	0.066	0.75	0.178
5	75	3	0.90	0.069	0.75	0.179
5	75	10	0.75	0.069	0.75	0.178
5	75	3	0.99	0.070	0.75	0.178
5	100	3	0.99	0.070	0.75	0.178
5	100	10	0.75	0.070	0.75	0.179
5	100	3	0.90	0.071	0.75	0.177
5	200	10	0.75	0.075	0.70	0.178
5	200	3	0.99	0.075	0.70	0.179
5	200	3	0.90	0.075	0.70	0.179

Table .13: Hyperparameter tuning results for S-learner in DGP 40

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
5	100	3	0.99	0.018	0.95	0.075
2	100	3	0.90	0.019	0.95	0.078
5	100	3	0.90	0.019	0.95	0.074
5	200	3	0.99	0.019	0.95	0.075
5	200	3	0.90	0.019	0.95	0.075
2	75	3	0.90	0.019	0.95	0.078
3	200	3	0.99	0.019	0.95	0.074
5	75	3	0.99	0.019	0.95	0.075
5	100	10	0.75	0.020	0.95	0.077
5	200	10	0.75	0.020	0.95	0.078
1	200	3	0.99	0.020	0.95	0.082
1	100	3	0.90	0.020	0.95	0.081
1	100	10	0.75	0.020	0.95	0.083
3	200	3	0.90	0.020	0.95	0.074
1	75	10	0.75	0.020	0.95	0.083
2	200	10	0.75	0.020	0.95	0.079
3	100	3	0.99	0.020	0.95	0.075
2	100	10	0.75	0.020	0.95	0.081
2	75	10	0.75	0.020	0.95	0.082
3	100	3	0.90	0.020	0.95	0.075
5	75	10	0.75	0.020	0.95	0.079
5	75	3	0.90	0.020	0.95	0.076
3	75	3	0.90	0.020	0.95	0.077
3	100	10	0.75	0.020	0.95	0.078
3	75	3	0.99	0.020	0.95	0.077
2	200	3	0.99	0.020	0.95	0.076
2	200	3	0.90	0.020	0.95	0.076
1	75	3	0.90	0.020	0.95	0.082
2	75	3	0.99	0.021	0.95	0.077
1	100	3	0.99	0.021	0.95	0.079
3	75	10	0.75	0.021	0.95	0.079
1	200	3	0.90	0.021	0.95	0.081
2	100	3	0.99	0.021	0.90	0.077
3	200	10	0.75	0.021	0.95	0.077
1	200	10	0.75	0.021	0.95	0.082
1	75	3	0.99	0.022	0.90	0.081

Table .14: Hyperparameter tuning results for S-learner in DGP 60

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
1	75	3	0.90	0.059	0.95	0.182
1	75	3	0.99	0.059	0.95	0.181
1	100	10	0.75	0.059	0.95	0.182
1	75	10	0.75	0.059	0.95	0.181
1	100	3	0.99	0.060	0.95	0.182
1	100	3	0.90	0.060	0.95	0.182
1	200	3	0.99	0.060	0.95	0.183
1	200	10	0.75	0.060	0.95	0.184
1	200	3	0.90	0.060	0.95	0.185
2	75	10	0.75	0.062	0.90	0.180
2	75	3	0.99	0.062	0.90	0.180
2	75	3	0.90	0.062	0.90	0.179
2	100	3	0.90	0.062	0.90	0.179
2	100	10	0.75	0.063	0.90	0.180
2	100	3	0.99	0.063	0.90	0.179
2	200	3	0.99	0.065	0.90	0.180
2	200	3	0.90	0.065	0.85	0.179
2	200	10	0.75	0.065	0.90	0.179
3	75	10	0.75	0.069	0.75	0.179
3	75	3	0.90	0.069	0.75	0.178
3	75	3	0.99	0.069	0.75	0.178
3	100	10	0.75	0.071	0.70	0.179
3	100	3	0.99	0.071	0.70	0.178
3	100	3	0.90	0.071	0.70	0.177
3	200	10	0.75	0.076	0.70	0.178
3	200	3	0.99	0.076	0.70	0.177
3	200	3	0.90	0.076	0.70	0.178
5	75	3	0.99	0.090	0.60	0.176
5	75	10	0.75	0.090	0.60	0.177
5	75	3	0.90	0.090	0.60	0.177
5	100	10	0.75	0.095	0.55	0.177
5	100	3	0.99	0.095	0.55	0.176
5	100	3	0.90	0.095	0.55	0.176
5	200	3	0.90	0.108	0.40	0.176
5	200	3	0.99	0.108	0.40	0.176
5	200	10	0.75	0.108	0.35	0.176

Table .15: Hyperparameter tuning results for S-learner in DGP 63

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
1	75	10	0.75	0.099	1.00	0.467
1	75	3	0.90	0.100	1.00	0.468
1	75	3	0.99	0.100	1.00	0.467
1	100	10	0.75	0.104	1.00	0.472
1	100	3	0.99	0.105	1.00	0.474
1	100	3	0.90	0.105	1.00	0.476
1	200	3	0.99	0.112	1.00	0.494
2	75	3	0.90	0.112	0.95	0.458
1	200	3	0.90	0.113	1.00	0.489
1	200	10	0.75	0.113	1.00	0.493
2	75	10	0.75	0.113	0.95	0.463
2	75	3	0.99	0.113	0.95	0.460
2	100	3	0.99	0.114	0.95	0.465
2	100	10	0.75	0.115	0.95	0.465
2	100	3	0.90	0.115	0.95	0.465
2	200	3	0.99	0.119	0.95	0.469
2	200	10	0.75	0.119	0.95	0.469
2	200	3	0.90	0.120	0.95	0.471
3	75	10	0.75	0.129	0.90	0.451
3	75	3	0.90	0.129	0.90	0.450
3	75	3	0.99	0.129	0.90	0.451
3	100	10	0.75	0.132	0.90	0.455
3	100	3	0.90	0.133	0.90	0.451
3	100	3	0.99	0.133	0.90	0.454
3	200	3	0.99	0.139	0.90	0.456
3	200	3	0.90	0.139	0.90	0.454
3	200	10	0.75	0.139	0.85	0.454
5	75	3	0.90	0.178	0.80	0.434
5	75	10	0.75	0.179	0.80	0.435
5	75	3	0.99	0.179	0.80	0.431
5	100	10	0.75	0.187	0.80	0.434
5	100	3	0.99	0.187	0.75	0.433
5	100	3	0.90	0.187	0.75	0.434
5	200	3	0.99	0.206	0.70	0.429
5	200	10	0.75	0.206	0.70	0.429
5	200	3	0.90	0.207	0.70	0.433

Table .16: Hyperparameter tuning results for T-learner in DGP 30

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
1	200	10	0.75	0.090	0.60	0.202
1	200	3	0.99	0.090	0.60	0.202
1	200	3	0.90	0.090	0.60	0.201
1	100	3	0.90	0.091	0.65	0.201
1	100	10	0.75	0.091	0.60	0.203
1	100	3	0.99	0.092	0.65	0.202
1	75	3	0.90	0.093	0.65	0.203
1	75	10	0.75	0.093	0.65	0.204
1	75	3	0.99	0.093	0.65	0.201
2	75	3	0.90	0.110	0.50	0.200
2	75	10	0.75	0.111	0.50	0.201
2	75	3	0.99	0.111	0.50	0.200
2	100	3	0.99	0.112	0.50	0.201
2	100	3	0.90	0.112	0.50	0.201
2	100	10	0.75	0.113	0.50	0.199
2	200	10	0.75	0.119	0.45	0.200
2	200	3	0.99	0.119	0.45	0.200
2	200	3	0.90	0.120	0.45	0.200
3	75	3	0.90	0.132	0.30	0.199
3	75	10	0.75	0.132	0.35	0.200
3	75	3	0.99	0.133	0.35	0.201
3	100	3	0.99	0.137	0.30	0.201
3	100	10	0.75	0.137	0.30	0.199
3	100	3	0.90	0.138	0.30	0.199
3	200	3	0.90	0.152	0.30	0.200
3	200	3	0.99	0.153	0.25	0.198
3	200	10	0.75	0.153	0.25	0.200
5	75	3	0.99	0.181	0.05	0.200
5	75	3	0.90	0.181	0.05	0.199
5	75	10	0.75	0.182	0.05	0.199
5	100	3	0.90	0.192	0.00	0.200
5	100	10	0.75	0.192	0.00	0.200
5	100	3	0.99	0.192	0.00	0.199
5	200	3	0.99	0.223	0.00	0.198
5	200	3	0.90	0.223	0.00	0.198
5	200	10	0.75	0.223	0.00	0.199

Table .17: Hyperparameter tuning results for T-learner in DGP 40

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
5	75	3	0.99	0.031	0.90	0.112
5	75	3	0.90	0.031	0.95	0.112
5	100	10	0.75	0.031	0.95	0.119
5	100	3	0.90	0.032	0.95	0.120
5	75	10	0.75	0.032	0.95	0.112
5	100	3	0.99	0.032	0.95	0.119
3	200	3	0.99	0.033	0.90	0.091
3	200	3	0.90	0.034	0.90	0.091
3	200	10	0.75	0.034	0.95	0.097
1	75	10	0.75	0.035	0.85	0.110
1	200	10	0.75	0.035	0.90	0.130
1	75	3	0.90	0.035	0.85	0.103
2	200	3	0.90	0.035	0.80	0.091
2	200	10	0.75	0.036	0.90	0.099
1	75	3	0.99	0.036	0.90	0.103
2	200	3	0.99	0.036	0.80	0.090
3	100	10	0.75	0.036	0.90	0.091
1	100	3	0.99	0.036	0.85	0.103
5	200	10	0.75	0.036	0.95	0.142
3	100	3	0.99	0.036	0.85	0.085
3	100	3	0.90	0.036	0.85	0.086
1	100	3	0.90	0.036	0.80	0.108
3	75	3	0.90	0.036	0.90	0.083
2	100	3	0.90	0.036	0.85	0.090
2	100	10	0.75	0.036	0.90	0.094
1	100	10	0.75	0.036	0.90	0.113
2	75	3	0.90	0.036	0.85	0.088
3	75	10	0.75	0.037	0.90	0.090
2	100	3	0.99	0.037	0.85	0.088
1	200	3	0.90	0.037	0.85	0.118
3	75	3	0.99	0.037	0.90	0.083
2	75	3	0.99	0.037	0.85	0.088
1	200	3	0.99	0.037	0.85	0.119
5	200	3	0.90	0.038	0.95	0.145
2	75	10	0.75	0.038	0.90	0.092
5	200	3	0.99	0.038	0.95	0.145

Table .18: Hyperparameter tuning results for T-learner in DGP 60

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
1	75	3	0.99	0.052	0.95	0.181
2	200	3	0.90	0.052	0.95	0.178
1	100	3	0.90	0.052	0.95	0.180
1	75	3	0.90	0.052	0.95	0.181
2	100	3	0.90	0.053	0.90	0.177
1	75	10	0.75	0.053	0.95	0.181
2	200	3	0.99	0.053	0.90	0.176
2	100	3	0.99	0.053	0.95	0.177
2	75	10	0.75	0.053	0.90	0.178
2	75	3	0.90	0.053	0.95	0.177
2	100	10	0.75	0.053	0.90	0.178
2	200	10	0.75	0.053	0.95	0.178
1	100	3	0.99	0.053	0.95	0.180
2	75	3	0.99	0.053	0.90	0.177
1	100	10	0.75	0.053	0.95	0.181
1	200	3	0.90	0.053	0.95	0.181
1	200	3	0.99	0.053	0.95	0.181
1	200	10	0.75	0.053	0.95	0.182
3	75	3	0.90	0.054	0.85	0.177
3	75	3	0.99	0.054	0.90	0.176
3	200	3	0.90	0.054	0.85	0.176
3	100	10	0.75	0.054	0.85	0.176
3	75	10	0.75	0.054	0.85	0.175
3	200	3	0.99	0.054	0.90	0.177
3	100	3	0.90	0.054	0.90	0.176
3	200	10	0.75	0.054	0.85	0.176
3	100	3	0.99	0.054	0.85	0.175
5	75	10	0.75	0.057	0.90	0.176
5	75	3	0.99	0.057	0.85	0.176
5	100	10	0.75	0.057	0.80	0.177
5	100	3	0.99	0.057	0.85	0.177
5	100	3	0.90	0.057	0.80	0.177
5	75	3	0.90	0.057	0.85	0.177
5	200	10	0.75	0.058	0.85	0.177
5	200	3	0.99	0.058	0.85	0.178
5	200	3	0.90	0.058	0.90	0.178

Table .19: Hyperparameter tuning results for T-learner in DGP 63

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
3	75	3	0.99	0.103	0.95	0.459
3	75	3	0.90	0.103	0.95	0.459
3	100	10	0.75	0.103	0.95	0.460
3	75	10	0.75	0.103	0.95	0.459
3	100	3	0.99	0.103	0.95	0.460
3	100	3	0.90	0.104	0.95	0.462
2	75	10	0.75	0.104	0.95	0.483
3	200	3	0.99	0.104	0.95	0.465
2	75	3	0.99	0.104	0.95	0.481
2	75	3	0.90	0.105	0.95	0.479
3	200	3	0.90	0.105	0.95	0.462
3	200	10	0.75	0.105	0.95	0.464
2	100	3	0.90	0.106	0.95	0.487
2	100	3	0.99	0.106	0.95	0.485
2	100	10	0.75	0.106	0.95	0.484
2	200	3	0.99	0.107	0.95	0.491
2	200	3	0.90	0.107	0.95	0.491
2	200	10	0.75	0.107	0.95	0.490
1	75	3	0.90	0.109	1.00	0.516
1	75	3	0.99	0.110	1.00	0.515
1	75	10	0.75	0.111	1.00	0.516
1	100	10	0.75	0.111	1.00	0.521
5	75	3	0.90	0.112	0.90	0.431
1	100	3	0.90	0.112	1.00	0.524
1	100	3	0.99	0.112	1.00	0.525
5	75	3	0.99	0.112	0.90	0.429
5	75	10	0.75	0.112	0.90	0.430
5	100	10	0.75	0.113	0.90	0.431
5	100	3	0.90	0.113	0.90	0.432
5	100	3	0.99	0.113	0.90	0.432
5	200	3	0.99	0.115	0.90	0.433
5	200	3	0.90	0.115	0.90	0.434
5	200	10	0.75	0.115	0.90	0.431
1	200	3	0.99	0.115	1.00	0.538
1	200	10	0.75	0.115	1.00	0.540
1	200	3	0.90	0.116	1.00	0.537

Table .20: Hyperparameter tuning results for PS-GLM in DGP 30

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
1	200	3	0.99	0.060	0.80	0.178
1	200	3	0.90	0.060	0.85	0.179
1	200	10	0.75	0.060	0.90	0.181
1	100	3	0.99	0.061	0.80	0.179
1	100	10	0.75	0.062	0.85	0.179
2	200	10	0.75	0.062	0.75	0.180
2	200	3	0.90	0.062	0.85	0.179
2	100	3	0.99	0.062	0.75	0.179
1	100	3	0.90	0.062	0.75	0.178
1	75	10	0.75	0.063	0.75	0.179
2	100	10	0.75	0.063	0.75	0.178
1	75	3	0.90	0.063	0.80	0.180
2	100	3	0.90	0.063	0.75	0.180
2	200	3	0.99	0.063	0.85	0.181
2	75	3	0.99	0.064	0.75	0.180
1	75	3	0.99	0.064	0.80	0.180
3	200	10	0.75	0.065	0.75	0.179
2	75	3	0.90	0.065	0.75	0.180
3	200	3	0.99	0.065	0.75	0.179
3	100	3	0.99	0.065	0.75	0.179
2	75	10	0.75	0.065	0.75	0.179
3	100	10	0.75	0.065	0.75	0.179
3	75	3	0.90	0.065	0.75	0.178
3	100	3	0.90	0.065	0.75	0.179
3	75	3	0.99	0.066	0.75	0.178
3	200	3	0.90	0.066	0.75	0.179
3	75	10	0.75	0.066	0.75	0.179
5	75	3	0.99	0.069	0.75	0.178
5	75	3	0.90	0.070	0.75	0.178
5	100	3	0.90	0.070	0.75	0.178
5	100	3	0.99	0.070	0.75	0.178
5	100	10	0.75	0.070	0.75	0.178
5	75	10	0.75	0.071	0.75	0.180
5	200	3	0.99	0.074	0.70	0.179
5	200	10	0.75	0.075	0.70	0.179
5	200	3	0.90	0.075	0.75	0.178

Table .21: Hyperparameter tuning results for PS-GLM in DGP 40

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
5	100	3	0.99	0.019	0.95	0.076
5	75	3	0.99	0.019	0.95	0.076
5	200	10	0.75	0.019	0.95	0.078
5	75	10	0.75	0.019	0.95	0.078
2	75	3	0.90	0.019	0.95	0.077
3	100	10	0.75	0.020	0.95	0.078
5	100	10	0.75	0.020	0.95	0.078
5	75	3	0.90	0.020	0.95	0.076
3	75	10	0.75	0.020	0.95	0.079
5	200	3	0.90	0.020	0.95	0.076
1	75	3	0.99	0.020	0.95	0.081
2	100	10	0.75	0.020	0.95	0.079
5	100	3	0.90	0.020	0.95	0.076
5	200	3	0.99	0.020	0.95	0.075
1	100	3	0.90	0.020	0.95	0.080
3	100	3	0.99	0.020	0.95	0.076
1	100	10	0.75	0.020	0.90	0.082
2	75	3	0.99	0.020	0.95	0.079
3	100	3	0.90	0.020	0.95	0.076
2	75	10	0.75	0.021	0.95	0.080
3	75	3	0.99	0.021	0.95	0.077
1	200	3	0.99	0.021	0.95	0.081
3	200	3	0.90	0.021	0.95	0.075
3	200	3	0.99	0.021	0.95	0.074
2	200	10	0.75	0.021	0.95	0.079
2	200	3	0.99	0.021	0.95	0.076
3	200	10	0.75	0.021	0.95	0.077
2	100	3	0.90	0.021	0.90	0.077
1	200	3	0.90	0.021	0.95	0.081
1	75	10	0.75	0.021	0.95	0.083
2	100	3	0.99	0.021	0.90	0.077
3	75	3	0.90	0.021	0.90	0.077
1	75	3	0.90	0.021	0.95	0.081
1	100	3	0.99	0.022	0.90	0.078
2	200	3	0.90	0.022	0.90	0.077
1	200	10	0.75	0.022	0.95	0.083

Table .22: Hyperparameter tuning results for PS-GLM in DGP 60

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
1	75	3	0.90	0.058	0.95	0.182
1	100	3	0.99	0.058	0.95	0.182
1	75	10	0.75	0.058	0.95	0.182
1	100	3	0.90	0.058	0.95	0.182
1	100	10	0.75	0.059	0.95	0.182
1	75	3	0.99	0.059	0.95	0.182
1	200	10	0.75	0.059	0.95	0.184
1	200	3	0.90	0.059	0.95	0.186
1	200	3	0.99	0.060	0.95	0.183
2	75	3	0.99	0.061	0.90	0.180
2	75	10	0.75	0.062	0.90	0.180
2	75	3	0.90	0.062	0.90	0.180
2	100	10	0.75	0.062	0.90	0.180
2	100	3	0.90	0.063	0.90	0.181
2	100	3	0.99	0.063	0.90	0.180
2	200	3	0.99	0.064	0.90	0.181
2	200	10	0.75	0.064	0.90	0.181
2	200	3	0.90	0.064	0.90	0.181
3	75	3	0.90	0.068	0.75	0.178
3	75	3	0.99	0.068	0.75	0.179
3	75	10	0.75	0.068	0.75	0.178
3	100	10	0.75	0.070	0.75	0.179
3	100	3	0.90	0.070	0.70	0.179
3	100	3	0.99	0.070	0.70	0.179
3	200	3	0.90	0.075	0.70	0.178
3	200	10	0.75	0.075	0.70	0.178
3	200	3	0.99	0.075	0.70	0.178
5	75	3	0.90	0.089	0.60	0.177
5	75	3	0.99	0.089	0.60	0.176
5	75	10	0.75	0.089	0.60	0.177
5	100	3	0.99	0.094	0.60	0.177
5	100	3	0.90	0.094	0.55	0.177
5	100	10	0.75	0.094	0.55	0.176
5	200	3	0.99	0.107	0.40	0.177
5	200	3	0.90	0.107	0.40	0.177
5	200	10	0.75	0.107	0.40	0.176

Table .23: Hyperparameter tuning results for PS-GLM in DGP 63

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
1	75	3	0.90	0.108	1.00	0.498
1	75	10	0.75	0.109	1.00	0.498
1	75	3	0.99	0.111	1.00	0.497
1	100	3	0.90	0.112	1.00	0.503
1	100	3	0.99	0.114	1.00	0.506
1	100	10	0.75	0.115	1.00	0.506
1	200	10	0.75	0.120	1.00	0.521
2	75	3	0.90	0.121	0.95	0.487
2	75	3	0.99	0.122	0.95	0.485
2	75	10	0.75	0.122	0.95	0.486
1	200	3	0.90	0.123	1.00	0.517
1	200	3	0.99	0.123	1.00	0.517
2	100	10	0.75	0.125	0.95	0.490
2	100	3	0.99	0.125	0.95	0.489
2	100	3	0.90	0.126	0.95	0.491
2	200	10	0.75	0.129	0.95	0.493
2	200	3	0.99	0.129	0.95	0.490
2	200	3	0.90	0.130	0.95	0.492
3	75	10	0.75	0.139	0.95	0.476
3	75	3	0.90	0.139	0.90	0.475
3	75	3	0.99	0.140	0.90	0.474
3	100	10	0.75	0.143	0.90	0.476
3	100	3	0.90	0.143	0.90	0.476
3	100	3	0.99	0.143	0.90	0.478
3	200	3	0.99	0.151	0.90	0.475
3	200	3	0.90	0.151	0.90	0.477
3	200	10	0.75	0.151	0.90	0.475
5	75	3	0.99	0.191	0.75	0.454
5	75	10	0.75	0.192	0.80	0.452
5	75	3	0.90	0.192	0.70	0.455
5	100	3	0.90	0.200	0.70	0.454
5	100	3	0.99	0.201	0.70	0.454
5	100	10	0.75	0.201	0.70	0.454
5	200	3	0.99	0.222	0.65	0.451
5	200	3	0.90	0.222	0.65	0.449
5	200	10	0.75	0.222	0.70	0.449

Table .24: Hyperparameter tuning results for PS-BART in DGP 30

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
3	100	10	0.75	0.047	0.95	0.183
3	200	10	0.75	0.047	0.95	0.184
3	200	3	0.99	0.047	0.95	0.185
1	200	3	0.99	0.047	1.00	0.187
2	75	3	0.99	0.047	0.95	0.186
5	100	3	0.99	0.047	0.95	0.181
3	100	3	0.99	0.047	0.95	0.182
3	200	3	0.90	0.048	0.95	0.185
2	200	3	0.99	0.048	0.95	0.186
3	75	10	0.75	0.048	0.95	0.182
1	200	10	0.75	0.048	0.95	0.187
2	75	3	0.90	0.048	0.95	0.184
5	75	3	0.99	0.048	0.95	0.181
5	75	10	0.75	0.048	0.95	0.182
2	75	10	0.75	0.048	0.95	0.184
1	100	3	0.99	0.048	0.95	0.185
5	100	10	0.75	0.049	0.95	0.181
5	75	3	0.90	0.049	0.95	0.182
1	200	3	0.90	0.049	0.95	0.187
2	200	3	0.90	0.049	0.95	0.186
1	75	3	0.90	0.049	0.95	0.184
3	100	3	0.90	0.049	0.95	0.184
2	200	10	0.75	0.049	0.95	0.187
5	100	3	0.90	0.049	0.95	0.181
2	100	3	0.90	0.049	0.95	0.185
1	100	10	0.75	0.049	0.95	0.183
3	75	3	0.90	0.049	0.95	0.184
5	200	10	0.75	0.049	0.95	0.182
1	75	3	0.99	0.049	0.95	0.185
2	100	3	0.99	0.050	0.95	0.187
5	200	3	0.90	0.050	0.95	0.183
1	75	10	0.75	0.050	0.95	0.185
5	200	3	0.99	0.050	0.95	0.183
1	100	3	0.90	0.050	0.95	0.186
3	75	3	0.99	0.050	0.95	0.185
2	100	10	0.75	0.050	0.95	0.185

Table .25: Hyperparameter tuning results for PS-BART in DGP 40

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
5	200	10	0.75	0.019	0.95	0.080
5	100	3	0.90	0.020	0.95	0.077
5	200	3	0.99	0.020	0.95	0.076
2	75	3	0.99	0.020	0.90	0.081
3	100	3	0.99	0.020	0.95	0.077
5	75	3	0.99	0.020	0.95	0.077
5	100	10	0.75	0.020	0.95	0.079
5	200	3	0.90	0.020	0.95	0.077
3	100	10	0.75	0.020	0.95	0.080
5	100	3	0.99	0.020	0.95	0.076
3	200	10	0.75	0.020	0.95	0.080
3	100	3	0.90	0.020	0.95	0.077
3	200	3	0.99	0.020	0.95	0.077
3	75	3	0.90	0.020	0.95	0.078
5	75	3	0.90	0.020	0.95	0.078
2	75	3	0.90	0.020	0.95	0.079
5	75	10	0.75	0.020	0.95	0.080
3	75	10	0.75	0.020	0.95	0.081
2	100	3	0.90	0.020	0.95	0.080
3	200	3	0.90	0.021	0.95	0.076
2	100	10	0.75	0.021	0.95	0.081
3	75	3	0.99	0.021	0.95	0.078
2	200	3	0.99	0.021	0.95	0.080
1	75	3	0.90	0.021	0.95	0.083
2	75	10	0.75	0.021	0.95	0.083
2	200	10	0.75	0.022	0.95	0.082
2	200	3	0.90	0.022	0.95	0.079
2	100	3	0.99	0.022	0.95	0.078
1	75	10	0.75	0.022	0.95	0.085
1	75	3	0.99	0.022	0.95	0.084
1	100	3	0.90	0.022	0.95	0.086
1	200	10	0.75	0.022	0.90	0.089
1	100	10	0.75	0.022	0.95	0.085
1	200	3	0.99	0.023	0.95	0.085
1	200	3	0.90	0.023	0.95	0.085
1	100	3	0.99	0.023	0.90	0.084

Table .26: Hyperparameter tuning results for PS-BART in DGP 60

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
2	75	10	0.75	0.060	0.95	0.184
2	75	3	0.99	0.060	0.95	0.184
1	75	10	0.75	0.060	0.90	0.193
2	75	3	0.90	0.061	0.95	0.185
1	75	3	0.99	0.061	0.95	0.195
2	100	10	0.75	0.061	0.95	0.185
2	100	3	0.99	0.062	0.95	0.185
2	100	3	0.90	0.062	0.90	0.186
1	100	3	0.90	0.062	0.95	0.200
1	100	10	0.75	0.063	0.95	0.201
1	75	3	0.90	0.063	0.90	0.195
2	200	10	0.75	0.063	0.90	0.186
2	200	3	0.90	0.064	0.90	0.187
1	200	10	0.75	0.064	0.95	0.209
1	100	3	0.99	0.064	0.90	0.200
2	200	3	0.99	0.064	0.90	0.186
1	200	3	0.99	0.064	0.95	0.210
1	200	3	0.90	0.065	0.95	0.209
3	75	3	0.90	0.067	0.80	0.182
3	75	3	0.99	0.067	0.80	0.181
3	75	10	0.75	0.067	0.80	0.180
3	100	10	0.75	0.068	0.75	0.180
3	100	3	0.99	0.069	0.75	0.181
3	100	3	0.90	0.069	0.75	0.182
3	200	10	0.75	0.074	0.75	0.181
3	200	3	0.90	0.074	0.75	0.181
3	200	3	0.99	0.074	0.75	0.180
5	75	10	0.75	0.088	0.60	0.178
5	75	3	0.90	0.089	0.60	0.179
5	75	3	0.99	0.089	0.60	0.178
5	100	3	0.99	0.093	0.60	0.178
5	100	10	0.75	0.093	0.60	0.178
5	100	3	0.90	0.093	0.60	0.178
5	200	3	0.99	0.106	0.40	0.177
5	200	10	0.75	0.106	0.40	0.177
5	200	3	0.90	0.107	0.40	0.177

Table .27: Hyperparameter tuning results for PS-BART in DGP 63

k	m	ν	q	Relative RMSE	Coverage	Relative CI Length
1	75	3	0.99	0.117	1.00	0.525
1	75	10	0.75	0.120	1.00	0.523
1	100	10	0.75	0.122	0.95	0.539
1	75	3	0.90	0.123	0.95	0.529
2	75	10	0.75	0.125	0.95	0.495
2	75	3	0.99	0.126	0.95	0.498
1	100	3	0.90	0.126	1.00	0.540
2	75	3	0.90	0.127	0.95	0.497
2	100	3	0.99	0.129	0.95	0.497
2	100	10	0.75	0.130	0.95	0.499
2	100	3	0.90	0.130	0.95	0.500
1	100	3	0.99	0.130	0.95	0.538
2	200	3	0.90	0.132	0.95	0.502
2	200	10	0.75	0.133	0.95	0.501
2	200	3	0.99	0.133	0.95	0.500
1	200	3	0.99	0.136	0.95	0.561
1	200	3	0.90	0.138	0.95	0.560
1	200	10	0.75	0.140	0.95	0.562
3	75	10	0.75	0.140	0.90	0.477
3	75	3	0.90	0.141	0.90	0.475
3	75	3	0.99	0.142	0.90	0.477
3	100	3	0.90	0.145	0.90	0.478
3	100	10	0.75	0.146	0.90	0.479
3	100	3	0.99	0.146	0.90	0.475
3	200	3	0.99	0.151	0.90	0.476
3	200	10	0.75	0.152	0.90	0.478
3	200	3	0.90	0.152	0.90	0.476
5	75	3	0.99	0.190	0.70	0.452
5	75	3	0.90	0.191	0.70	0.451
5	75	10	0.75	0.191	0.70	0.452
5	100	10	0.75	0.199	0.70	0.450
5	100	3	0.90	0.199	0.70	0.451
5	100	3	0.99	0.200	0.70	0.448
5	200	10	0.75	0.218	0.65	0.446
5	200	3	0.90	0.219	0.65	0.445
5	200	3	0.99	0.220	0.65	0.446

Table .28: Hyperparameter tuning results for Causal Forest in DGP 30

Min Node Size	Sample Fraction	mtry	Relative RMSE	Coverage	Relative CI Length
20	0.50	26	0.096	0.55	0.194
10	0.50	26	0.098	0.55	0.194
50	0.50	26	0.099	0.55	0.194
5	0.50	26	0.100	0.55	0.193
20	0.30	26	0.130	0.30	0.189
50	0.30	26	0.131	0.25	0.189
5	0.30	26	0.133	0.25	0.189
10	0.30	26	0.133	0.30	0.190
10	0.50	13	0.144	0.15	0.183
50	0.50	13	0.145	0.15	0.182
20	0.50	13	0.147	0.15	0.183
5	0.50	13	0.147	0.10	0.183
50	0.30	13	0.195	0.00	0.180
10	0.30	13	0.196	0.00	0.180
20	0.30	13	0.197	0.00	0.180
5	0.30	13	0.198	0.00	0.180
50	0.50	9	0.222	0.00	0.177
5	0.50	9	0.223	0.00	0.177
20	0.50	9	0.223	0.00	0.177
10	0.50	9	0.224	0.00	0.177
50	0.30	9	0.284	0.00	0.176
20	0.30	9	0.285	0.00	0.175
5	0.30	9	0.285	0.00	0.176
10	0.30	9	0.285	0.00	0.176

Table .29: Hyperparameter tuning results for Causal Forest in DGP 40

Min Node Size	Sample Fraction	mtry	Relative RMSE	Coverage	Relative CI Length
5	0.50	22	0.055	1.00	0.228
50	0.50	22	0.057	1.00	0.231
5	0.50	11	0.060	1.00	0.242
20	0.50	22	0.061	1.00	0.234
10	0.50	11	0.063	1.00	0.241
20	0.50	11	0.064	1.00	0.241
10	0.50	22	0.064	1.00	0.234
50	0.50	11	0.064	1.00	0.243
50	0.50	8	0.067	1.00	0.248
10	0.50	8	0.068	1.00	0.252
10	0.30	22	0.069	1.00	0.252
5	0.50	8	0.069	1.00	0.248
5	0.30	22	0.069	1.00	0.253
20	0.50	8	0.071	1.00	0.254
20	0.30	22	0.071	1.00	0.251
50	0.30	22	0.072	1.00	0.253
5	0.30	11	0.074	1.00	0.262
10	0.30	11	0.075	1.00	0.261
20	0.30	11	0.076	1.00	0.261
20	0.30	8	0.077	1.00	0.266
10	0.30	8	0.077	1.00	0.267
50	0.30	11	0.078	1.00	0.264
5	0.30	8	0.078	1.00	0.268
50	0.30	8	0.080	1.00	0.268

Table .30: Hyperparameter tuning results for Causal Forest in DGP 60

Min Node Size	Sample Fraction	mtry	Relative RMSE	Coverage	Relative CI Length
5	0.50	11	0.060	0.85	0.180
5	0.30	11	0.061	0.80	0.181
20	0.30	11	0.061	0.85	0.181
10	0.50	11	0.061	0.85	0.180
50	0.50	11	0.061	0.85	0.180
10	0.30	11	0.061	0.80	0.181
20	0.50	11	0.062	0.85	0.180
10	0.50	16	0.062	0.85	0.179
50	0.50	16	0.062	0.85	0.179
5	0.30	16	0.062	0.85	0.180
5	0.50	16	0.063	0.85	0.180
20	0.50	16	0.063	0.85	0.179
10	0.30	16	0.063	0.85	0.180
50	0.30	11	0.063	0.80	0.180
50	0.30	16	0.063	0.85	0.180
5	0.50	26	0.063	0.85	0.179
20	0.30	16	0.063	0.80	0.180
50	0.50	26	0.064	0.85	0.179
20	0.50	26	0.064	0.85	0.179
10	0.50	26	0.064	0.85	0.179
5	0.30	26	0.064	0.85	0.180
20	0.30	26	0.065	0.80	0.179
50	0.30	26	0.065	0.85	0.180
10	0.30	26	0.065	0.85	0.179

Table .31: Hyperparameter tuning results for Causal Forest in DGP 63

Min Node Size	Sample Fraction	mtry	Relative RMSE	Coverage	Relative CI Length
50	0.30	10	0.121	0.80	0.367
20	0.30	10	0.122	0.80	0.367
10	0.30	10	0.126	0.80	0.367
5	0.30	10	0.126	0.80	0.368
10	0.30	15	0.127	0.80	0.382
10	0.50	10	0.127	0.75	0.383
50	0.50	10	0.128	0.80	0.384
5	0.30	15	0.128	0.80	0.383
20	0.30	15	0.129	0.80	0.383
50	0.30	15	0.130	0.75	0.382
20	0.50	10	0.131	0.80	0.384
5	0.50	10	0.132	0.80	0.385
20	0.30	26	0.133	0.85	0.405
10	0.50	15	0.133	0.80	0.405
10	0.30	26	0.134	0.90	0.405
20	0.50	15	0.135	0.85	0.403
5	0.50	15	0.136	0.80	0.405
50	0.30	26	0.136	0.85	0.404
50	0.50	15	0.136	0.80	0.406
5	0.30	26	0.137	0.85	0.404
10	0.50	26	0.138	0.90	0.431
20	0.50	26	0.139	0.90	0.431
50	0.50	26	0.141	0.85	0.431
5	0.50	26	0.144	0.80	0.431

Hiermit erkläre ich an Eides statt, dass ich die vorliegende Arbeit selbständig und ohne unerlaubte fremde Hilfe angefertigt, andere als die angegebenen Quellen und Hilfsmittel nicht benutzt und die den benutzten Quellen und Hilfsmitteln wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe.

Bamberg, den xx. Mai 2099

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(Unterschrift des Verfassers)